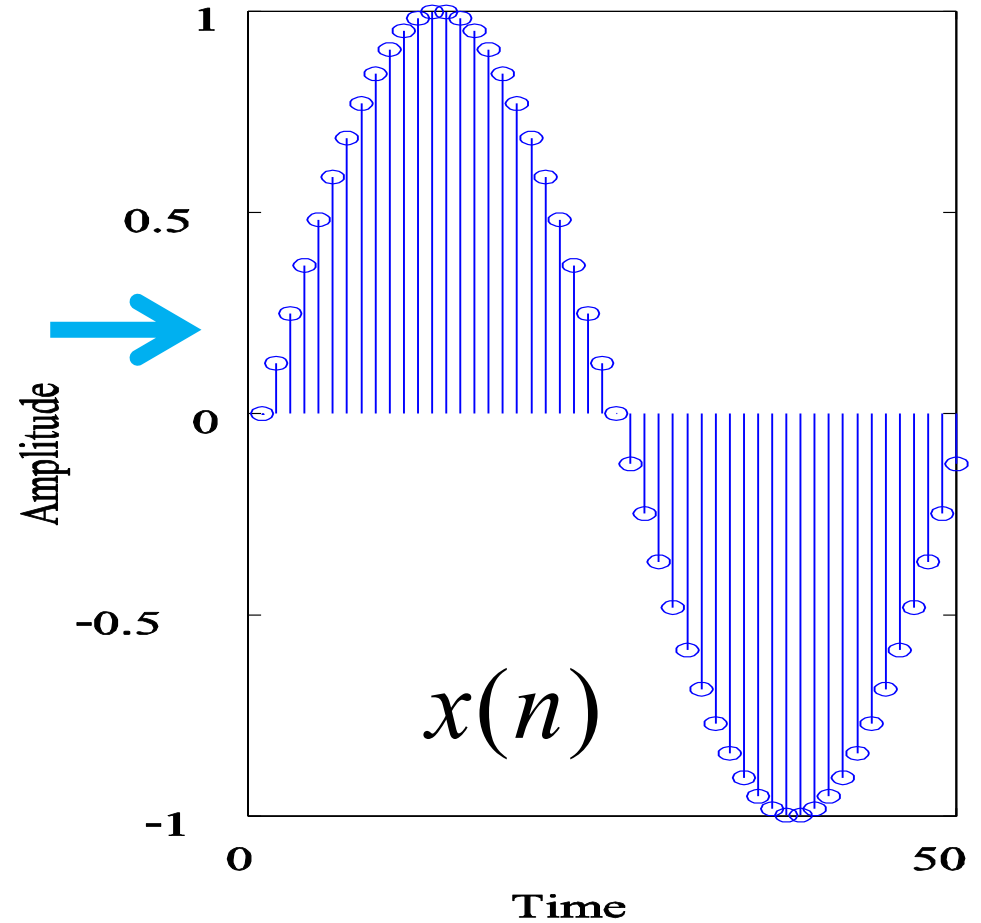
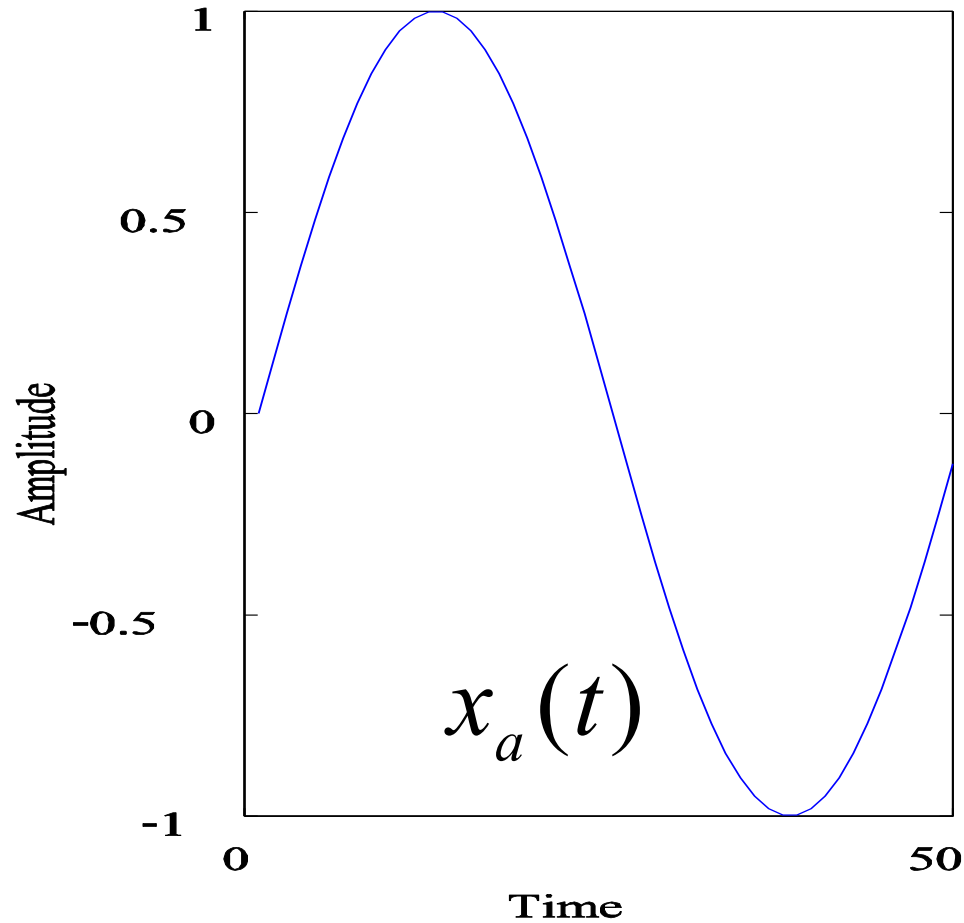


3

**The Discrete-Time
Fourier Analysis and Transform**

Continuous to Discrete-time Signals



Continuous-time Sinusoidal Signal

- Let us define the function of the Continuous-time sinusoidal signal as

$$x_a(t) = A \cos(2\pi ft + \theta) \quad t = \text{time(sec)}$$

- or $f = \text{Frequency(Hz)}$

$$x_a(t) = A \cos(\Omega t + \theta) \quad \theta = \text{Phase(radians)}$$

$\Omega = \text{Angular Frequency (radians/second)}$

Converting Continuous-time to Discrete-time Sinusoidal Signals

$$x_a(t)|_{t=nT} = A \cos(\Omega t + \theta)|_{t=nT}$$

$$= A \cos(\Omega nT + \theta)$$

$$= A \cos((\Omega T)n + \theta)$$

ω = Digital Frequency

(radians/sample)

$$= A \cos(\omega n + \theta)$$

$$\omega = \Omega T \quad = x(n)$$

Discrete-time Sinusoidal Signal

- The discrete-time signal is

$$x(n) = A \cos(\omega n + \theta)$$

n = sample

$\omega = \Omega T$ = Digital Frequency (radians/sample)

θ = Phase(radians)

Discrete-time Sinusoidal Signals

- Digital Angular frequency of Digital frequency is related to the analogue frequency f_a as

$$\begin{aligned}\omega &= \Omega T \\ &= 2\pi f_a T \\ &= 2\pi \frac{f_a}{f_s}\end{aligned}$$

- Where f_s is the sampling frequency

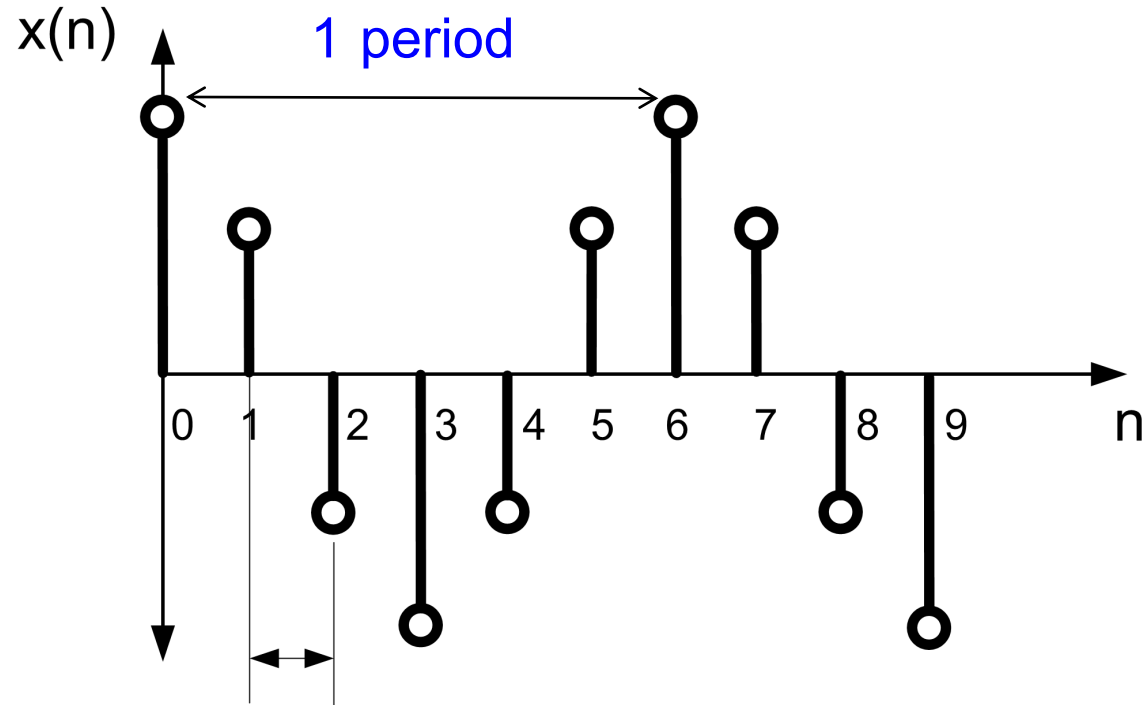
Normalised Frequency

- Define $f = \frac{f_a}{f_s}$, then

$$\omega = 2\pi f$$

- Where f is **Normalised frequency**: i.e. Normalised f_a with respect to the sampling frequency f_s .

$$x(n) = \cos\left(\frac{\pi}{3}n\right)$$



$$\omega_0 = \frac{\pi}{3} \text{ radians per sample}$$

$$f_0 = \frac{1}{6} \text{ cycles per sample}$$

Digital Frequency

- It is obvious that, for this signal, there are six samples per cycle.
- One cycle has 2π radians.
- Then each sample is separated by

$$\omega = \frac{2\pi}{6} \quad \text{radians/sample}$$

- or

$$\omega = \frac{\pi}{3} \quad \text{radians/sample}$$

$$x(n) = \cos(\omega_0 n)$$

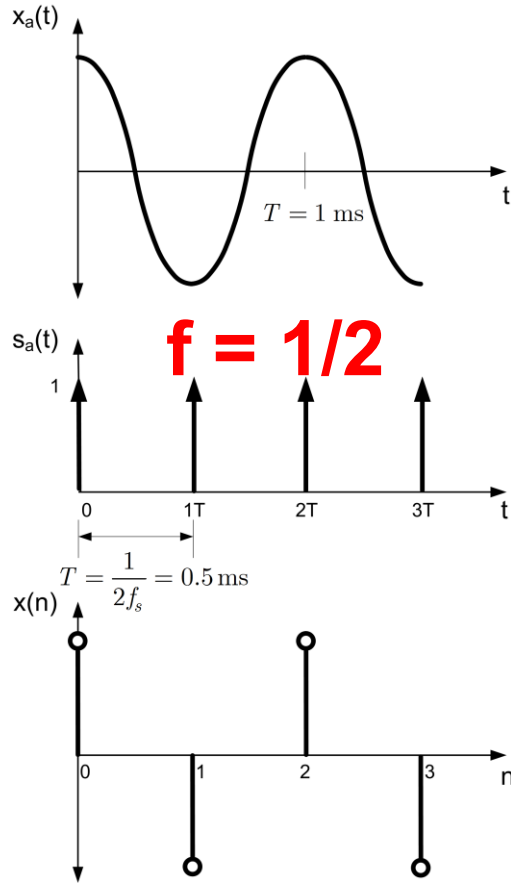
n	$\omega n = \frac{2\pi n}{6}$
0	$= 2\pi \left(\frac{0}{6}\right)$
1	$= 2\pi \left(\frac{1}{6}\right)$
2	$= 2\pi \left(\frac{2}{6}\right) = 2\pi \left(\frac{1}{3}\right)$
3	$= 2\pi \left(\frac{3}{6}\right) = \pi$
4	$= 2\pi \left(\frac{4}{6}\right) = 2\pi \left(\frac{2}{3}\right)$
5	$= 2\pi \left(\frac{5}{6}\right)$
6	$= 2\pi \left(\frac{6}{6}\right) = 2\pi$

N is the number of samples in 1 cycle

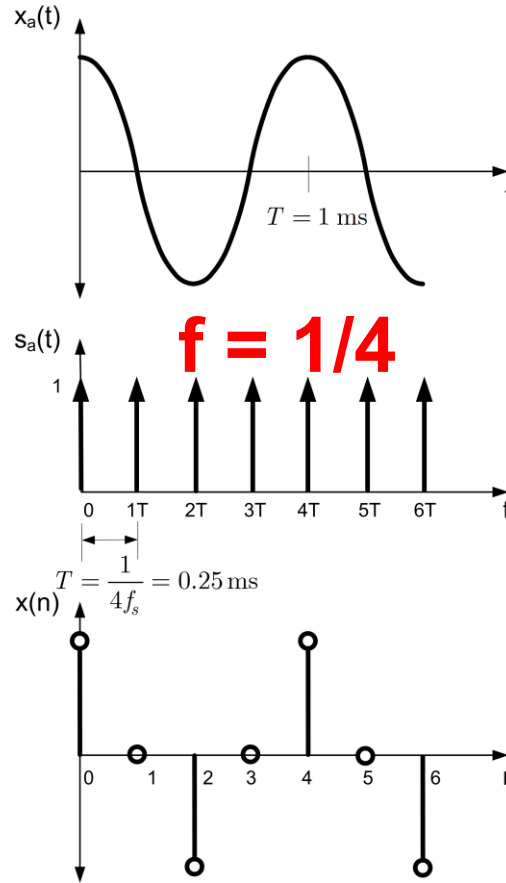
- From $\omega n = \frac{2\pi}{6} n$
- then $\omega = \frac{2\pi}{6}$ radians/sample
- Let N be the number of samples in 1 cycle, the digital frequency is defined as $\omega = \frac{2\pi}{N}$ radians/sample

When $f = 1/2, 1/4, 1/8$ cycles per sample

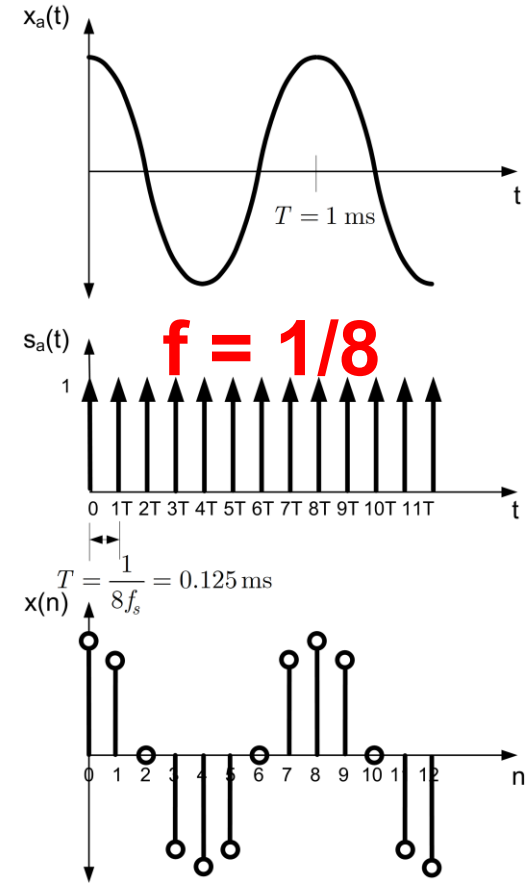
$f_a = 1$ kHz



$f_a = 1$ kHz



$f_a = 1$ kHz



Periodic Signals

- For a signal with period N , its value at the sample $n+N$ is equal to its value at sample n .

$$x(n + N) = x(n)$$

- A signal is periodic iff **frequency (f)* period (N) must be an integer k**

$$fN = k$$

- Otherwise it is called **aperiodic**.

1) For a periodic signal, k and N must be integer

- The interpretation is that the ratio $f = \frac{k}{N}$ must be a rational number, i.e. k and N must be integer numbers.

- Example

$$f = \frac{1}{5}$$

Rational

$$f = \frac{1}{\sqrt{2}}$$

Irrational

Proof

- Let $x(n)$ and $x(N+n)$ respectively be

$$x(n) = A \cos(2\pi f n + \theta)$$

$$x(N + n) = A \cos(2\pi f(N + n) + \theta)$$

For $f \cdot N$ is an integer

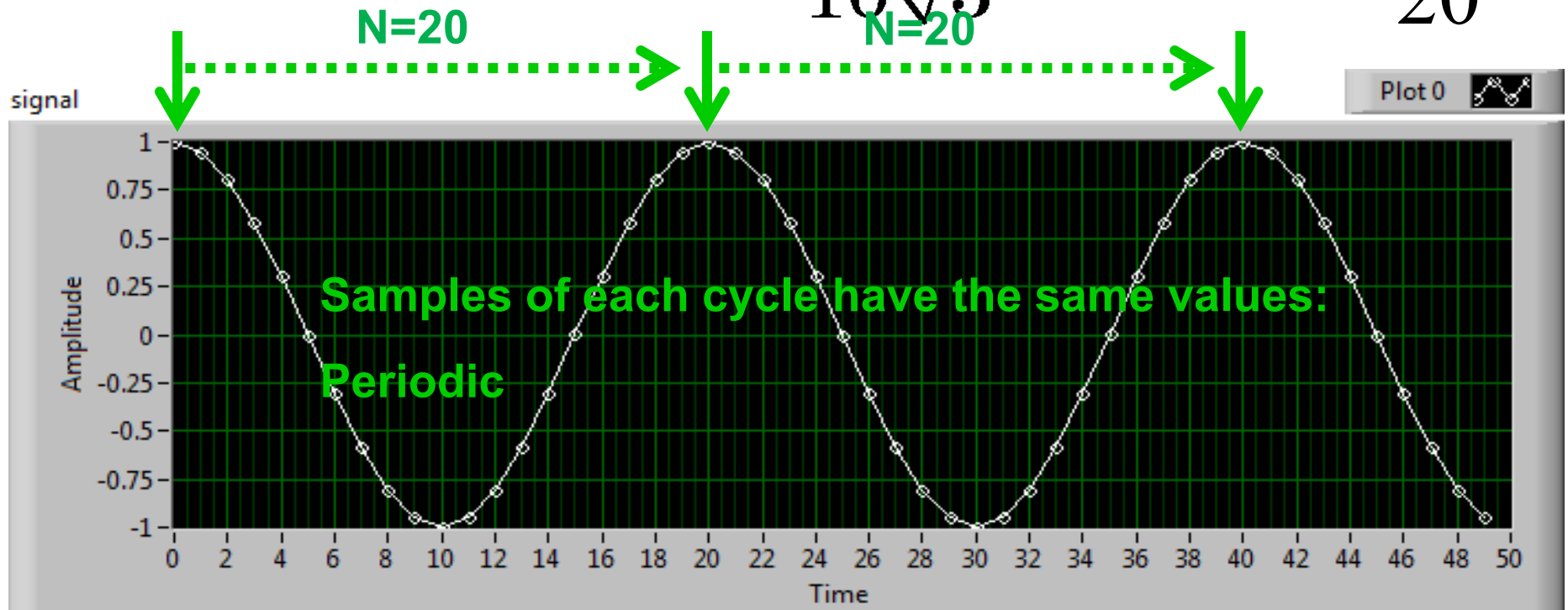
$$\begin{aligned}x(N + n) &= \cos(2\pi f(N + n) + \theta) \\&= \cos(2\pi fN + (2\pi fn + \theta)) \\&= \cos(2\pi fN) \cos(2\pi fn + \theta) - \sin(2\pi fN) \sin(2\pi fn + \theta) \\&\quad \uparrow \qquad \qquad \qquad \uparrow \\&\quad \cos(2\pi fN) = 1 \qquad \qquad \sin(2\pi fN) = 0\end{aligned}$$

When $f \cdot N$ is an integer, this term is 1

$$x(N + n) = x(n) \qquad \text{Q.E.D.}$$

$$x(n + N) = x(n)$$

Example 1 $x(n) = \cos\left(\frac{\sqrt{3}}{10\sqrt{3}} \pi n\right) = \cos\left(\frac{2\pi}{20} n\right)$



$$f = \frac{k}{N}$$

- For $\cos(\frac{2\pi}{20}n)$

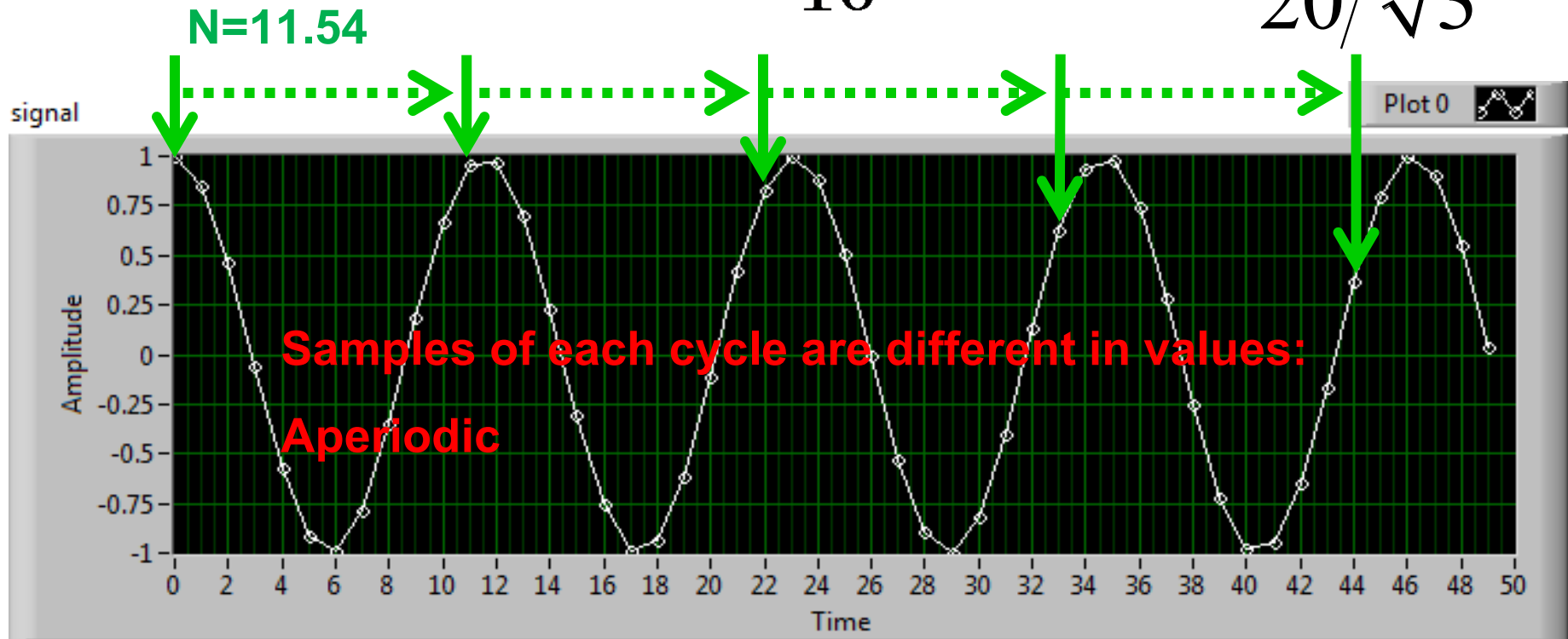
$$2\pi f = \frac{2\pi}{20} = 2\pi \frac{1}{20}$$

$$\therefore \frac{f}{N} = \frac{1}{20} \quad \text{Rational}$$

- So this signal is **periodic**.

$$x(n + N) \neq x(n)$$

- Example 2 $x(n) = \cos\left(\frac{\sqrt{3}}{10}\pi n\right) = \cos\left(\frac{2\pi}{20/\sqrt{3}}n\right)$



$$f = \frac{k}{N}$$

- For $\cos\left(\frac{2\pi}{20/\sqrt{3}}n\right)$

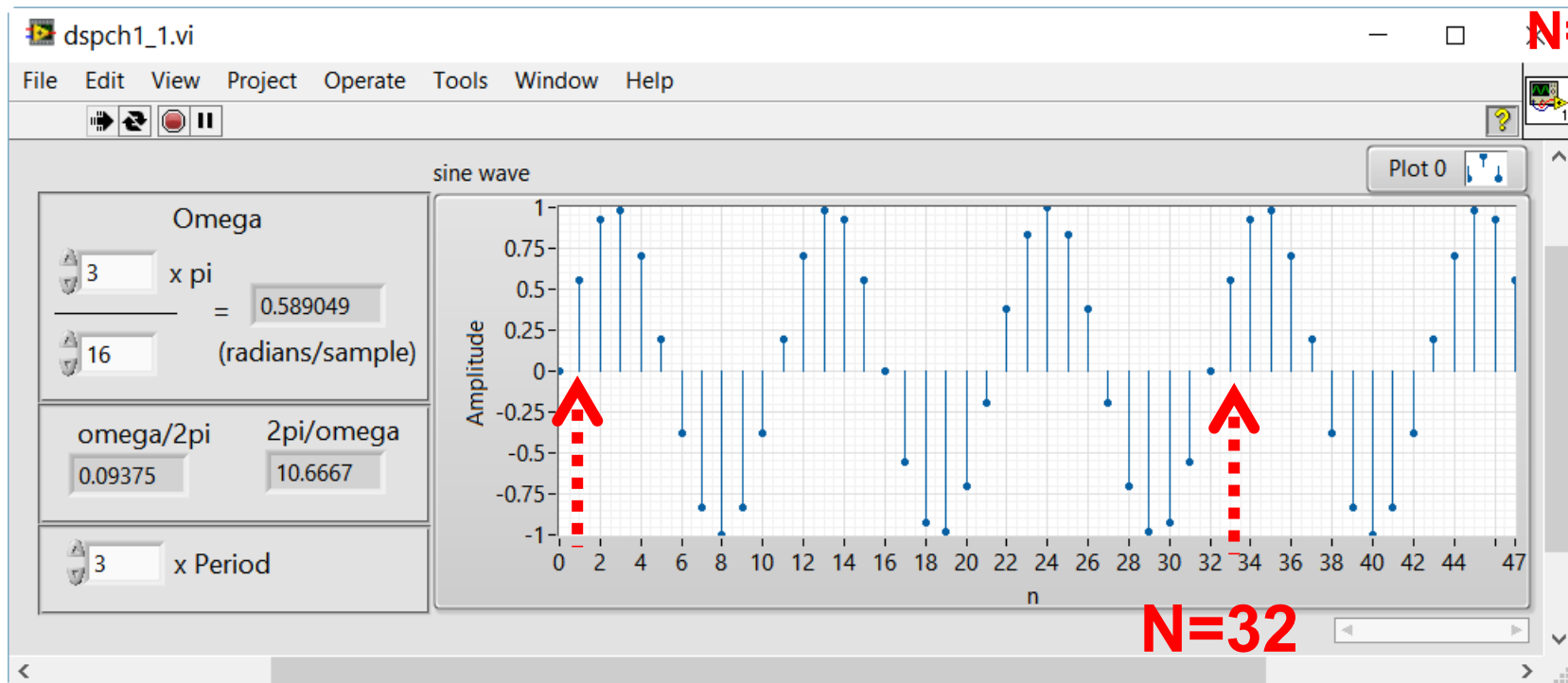
$$2\pi \frac{f}{N} = \frac{2\pi}{20/\sqrt{3}} = 2\pi \frac{\sqrt{3}}{20}$$

$$\therefore \frac{f}{N} = \frac{\sqrt{3}}{20} \quad \text{Irrational}$$

- So this signal is **aperiodic**.

$$\omega = \frac{3\pi}{16}$$

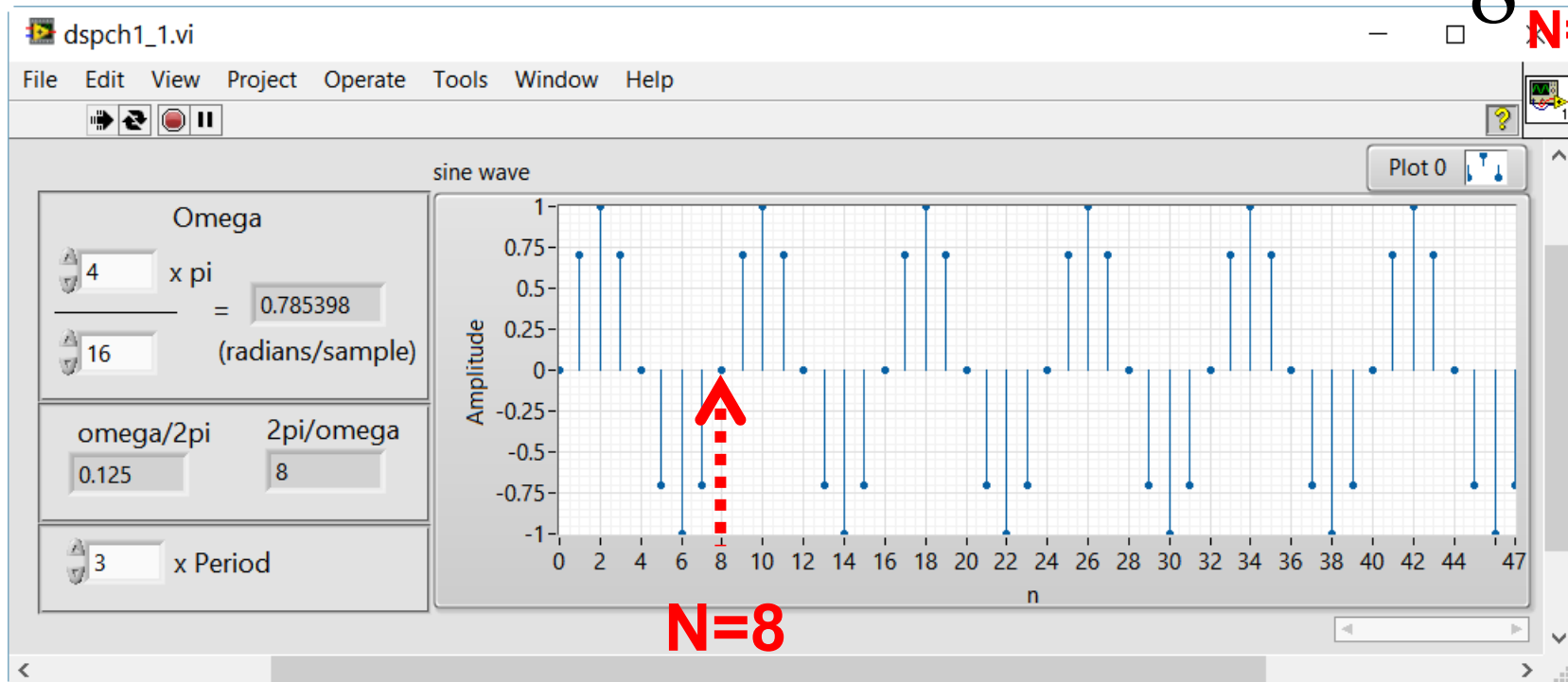
$$\omega = \frac{2 \times 3\pi}{2 \times 16} = \frac{6\pi}{32}$$



$$\omega = \frac{4\pi}{16}$$

$$\omega = \frac{2\pi}{8}$$

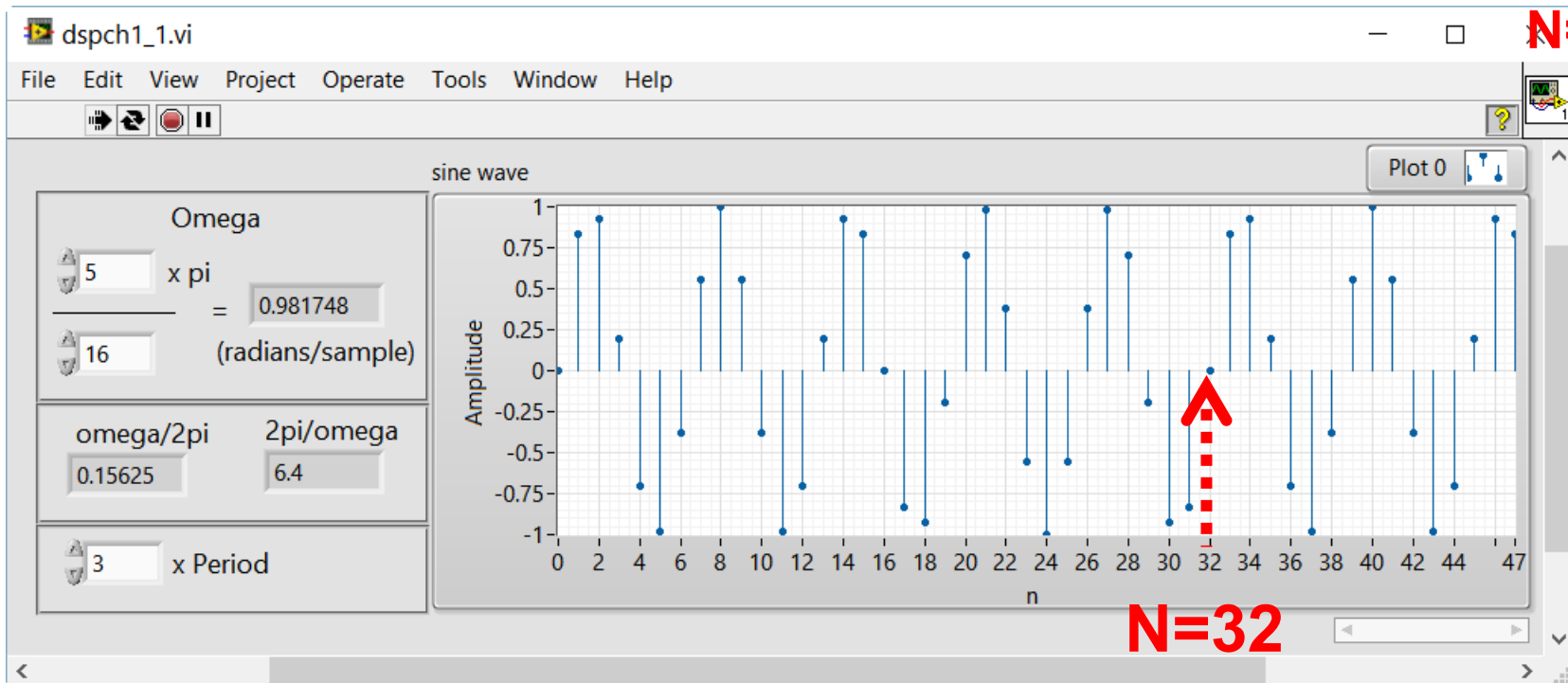
N=8



N=8

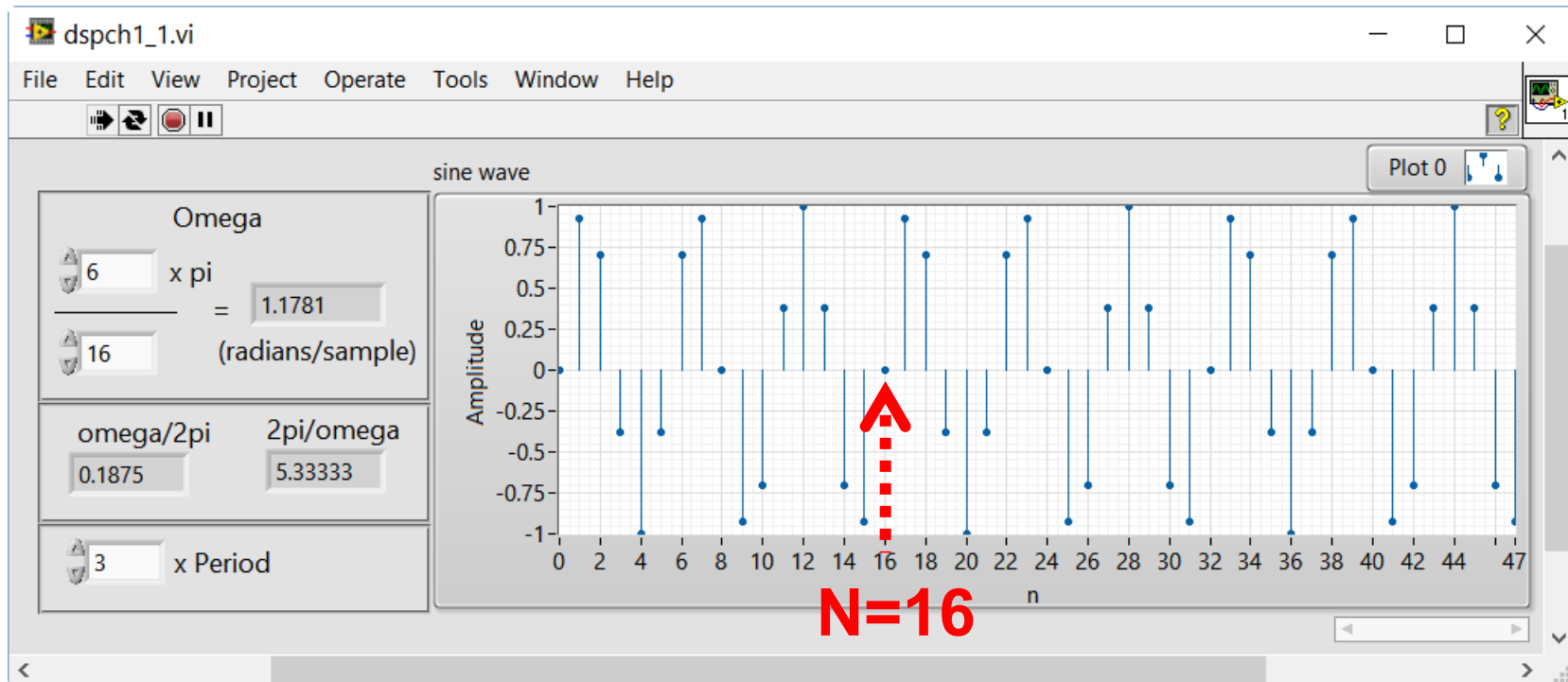
$$\omega = \frac{5\pi}{16}$$

$$\omega = \frac{2 \times 5\pi}{2 \times 16} = \frac{10\pi}{32}$$

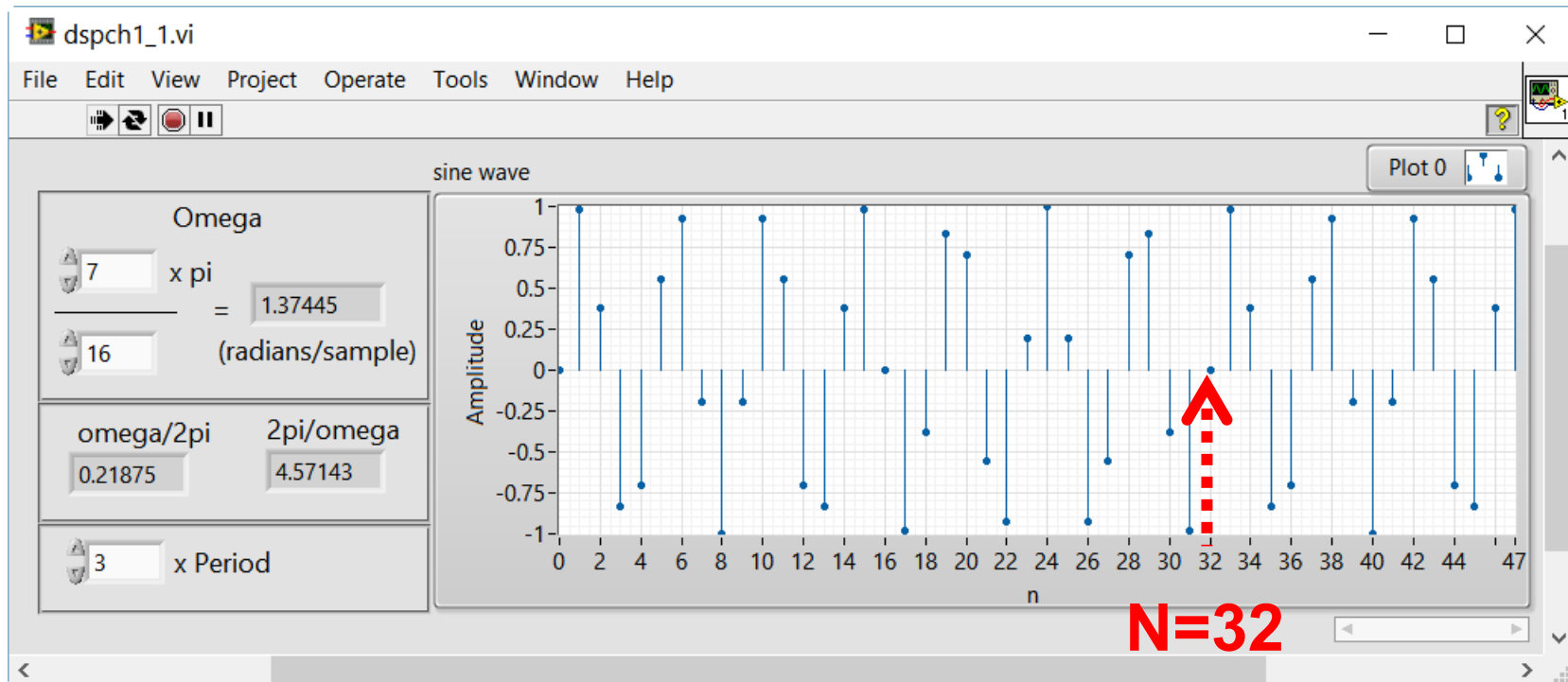


$$\omega = \frac{6\pi}{16}$$

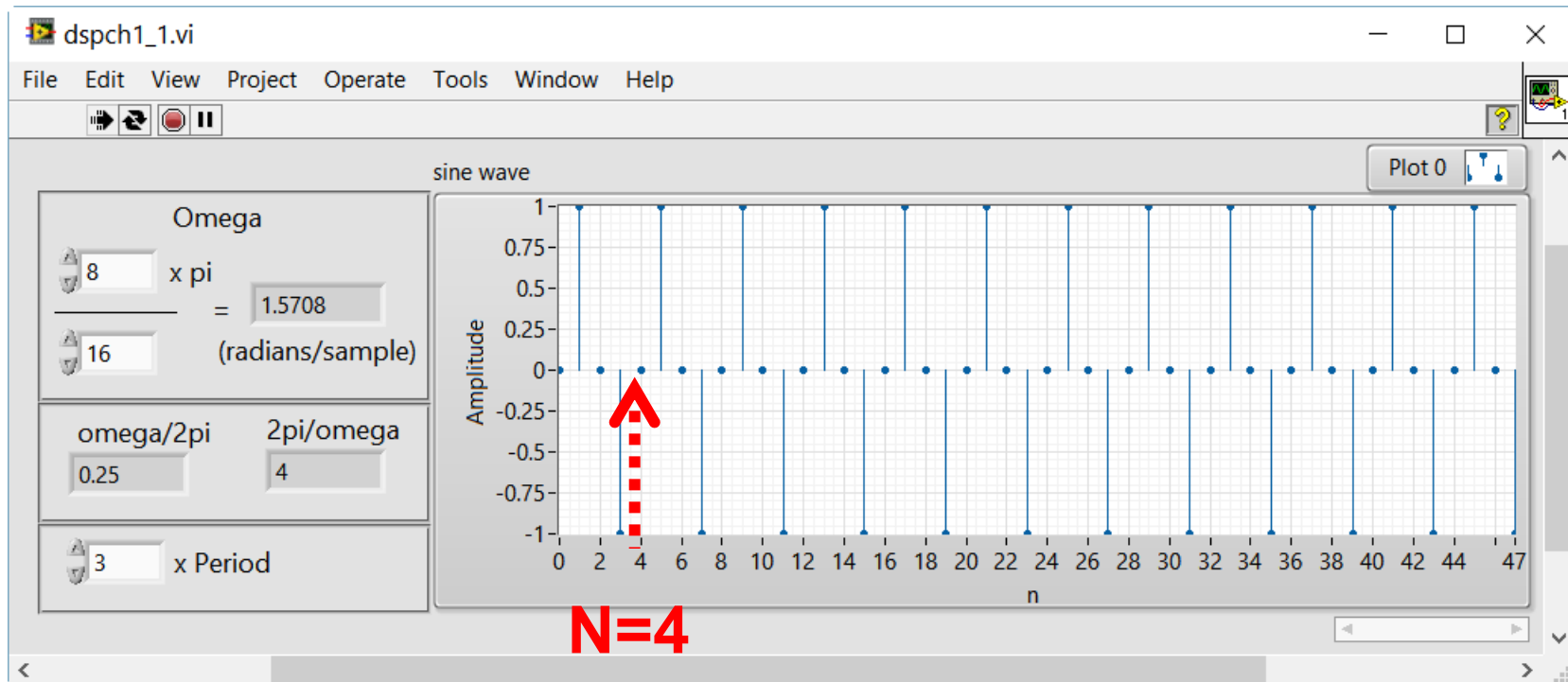
$$\omega = \frac{6\pi}{16} = \frac{3 \times 2\pi}{16}$$



$$\omega = \frac{7\pi}{16}$$



$$\omega = \frac{8\pi}{16}$$



Example 3.1

- Determine whether the following item is periodic or not and if it is what is its period N ?

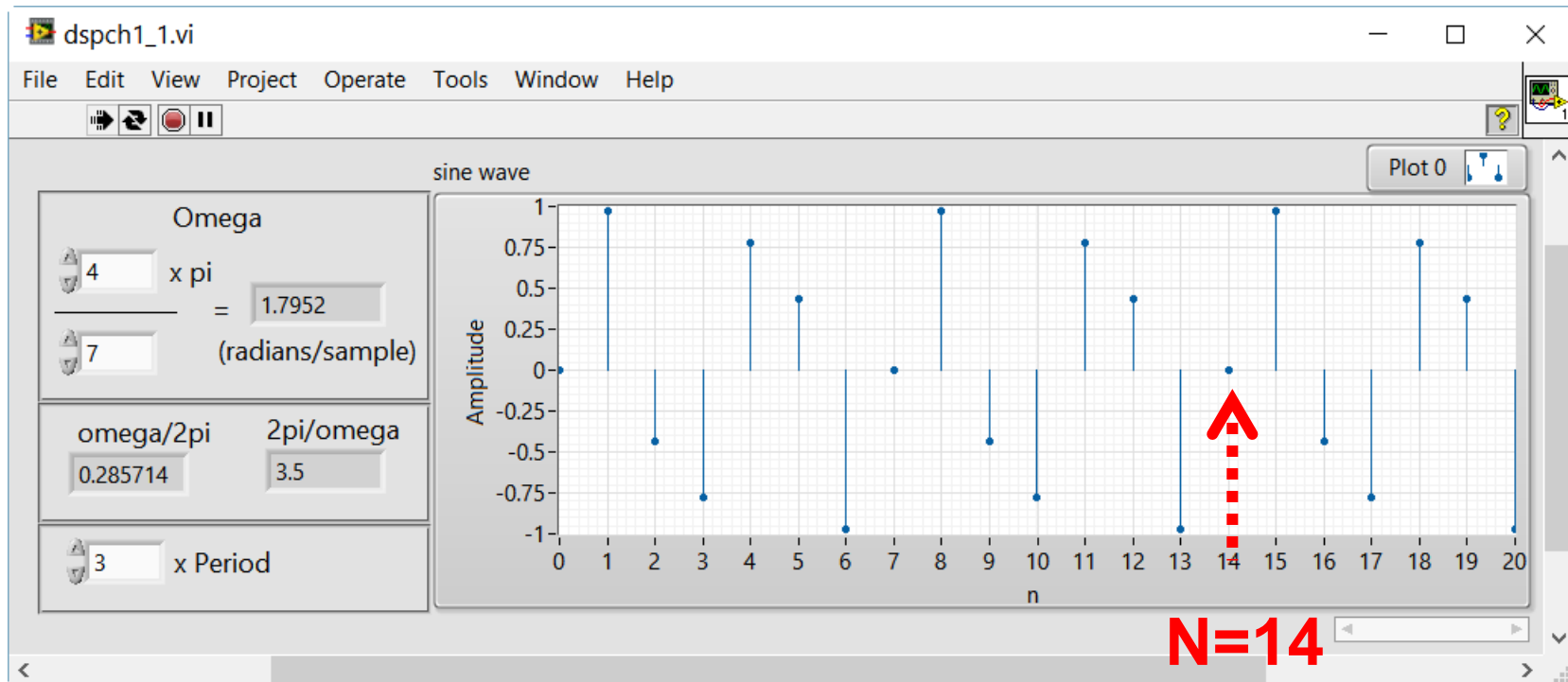
$$a) \quad x(n) = \cos\left(\frac{4\pi n}{7}\right)$$

$$b) \quad x(n) = \cos\left(\frac{13\pi n}{12}\right)$$

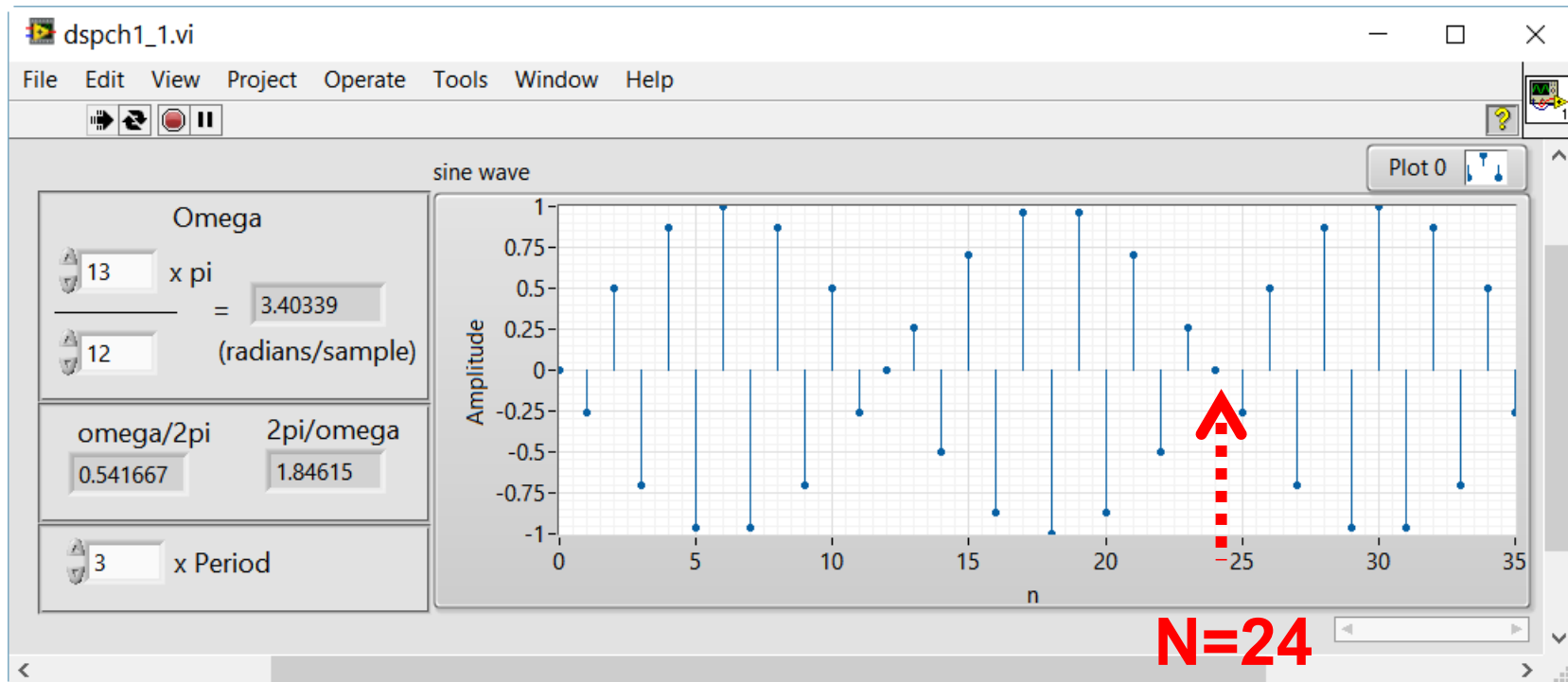
$$c) \quad x(n) = \cos\left(\frac{5\pi n}{9}\right)$$

$$d) \quad x(n) = \cos\left(\frac{\sqrt{5}\pi n}{10}\right)$$

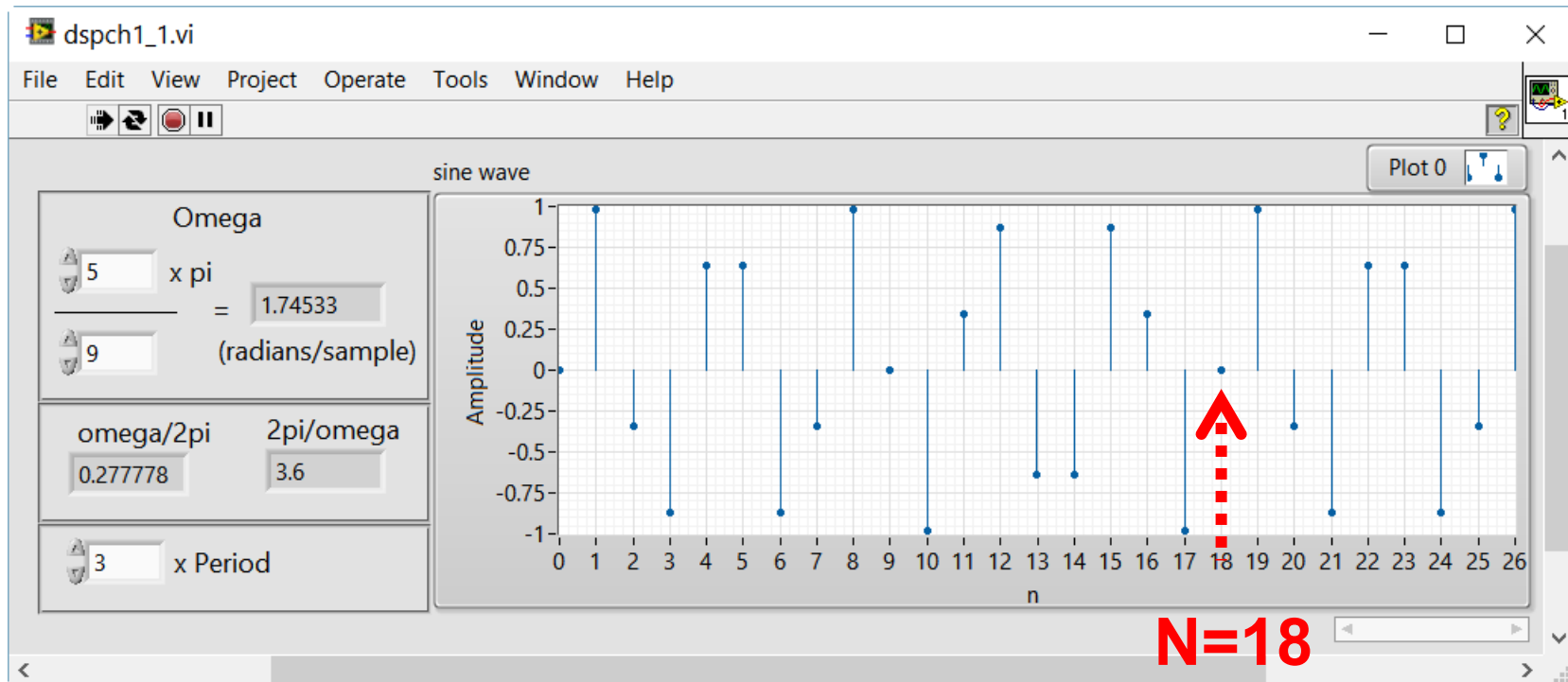
$$a) \quad x(n) = \cos\left(\frac{4\pi n}{7}\right)$$



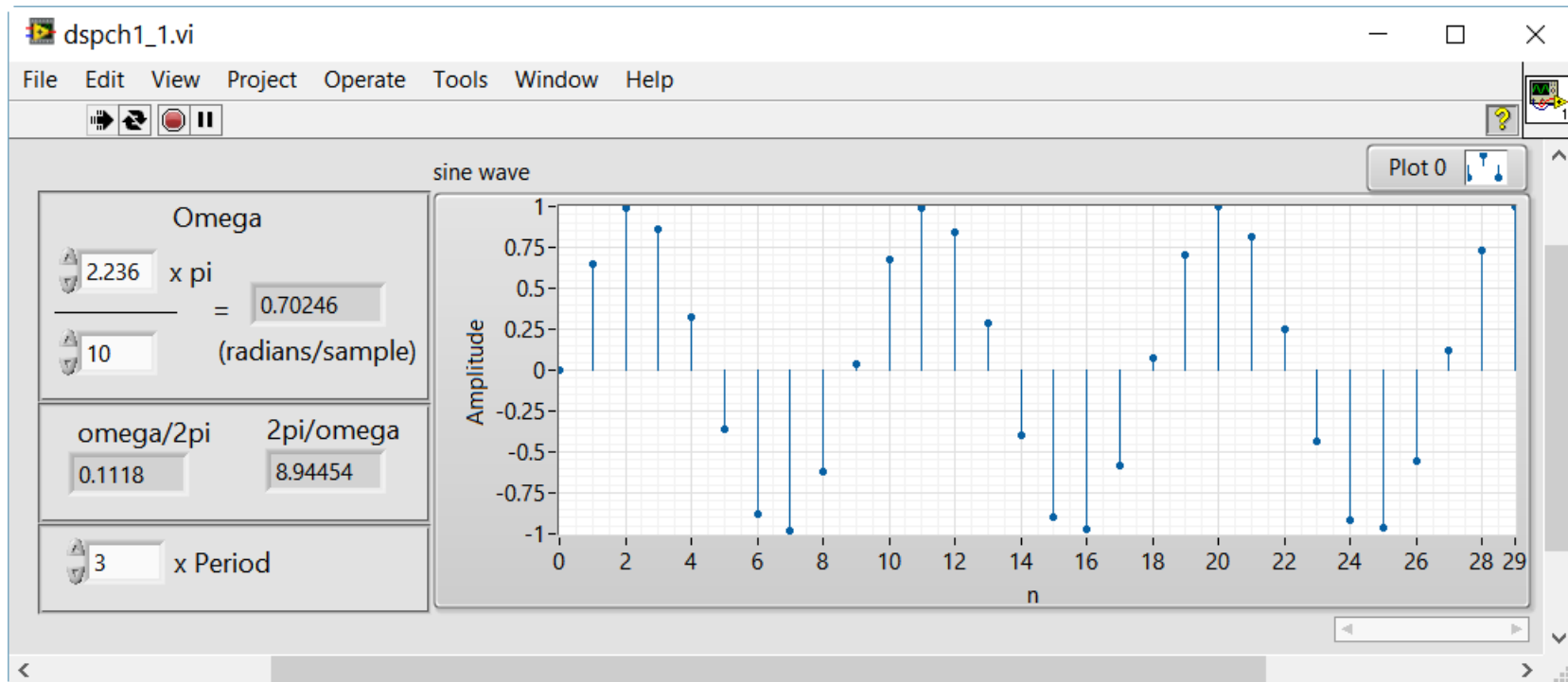
$$b) \quad x(n) = \cos\left(\frac{13\pi n}{12}\right)$$



$$c) \quad x(n) = \cos\left(\frac{5\pi n}{9}\right)$$



$$d) \quad x(n) = \cos\left(\frac{\sqrt{5}\pi n}{10}\right)$$



2 The maximum frequency is π radians

- For $x(n) = \cos(\omega n)$
- We can also write

$$\begin{aligned}x'(n) &= \cos(\omega' n) \\&= \cos((2\pi - \omega)n) \\&= x(n)\end{aligned}$$

- Where $\omega' = 2\pi - \omega$

Frequency

- We say that the frequency is **meaningful** when

$$0 \leq \omega \leq \pi$$

- Beyond that, the value of the cosine is repetitive
- Or we can say the range of frequency is

$$|\omega| \leq \pi$$

$$x(n) = \cos\left(\frac{\pi}{4}n\right)$$

$$\cos\left(\frac{\pi}{4} \cdot 3\right) = -\frac{\sqrt{2}}{2}$$

$$\cos\left(\frac{\pi}{4} \cdot 4\right) = -1$$

$$\cos\left(\frac{\pi}{4} \cdot 5\right) = -\frac{\sqrt{2}}{2}$$

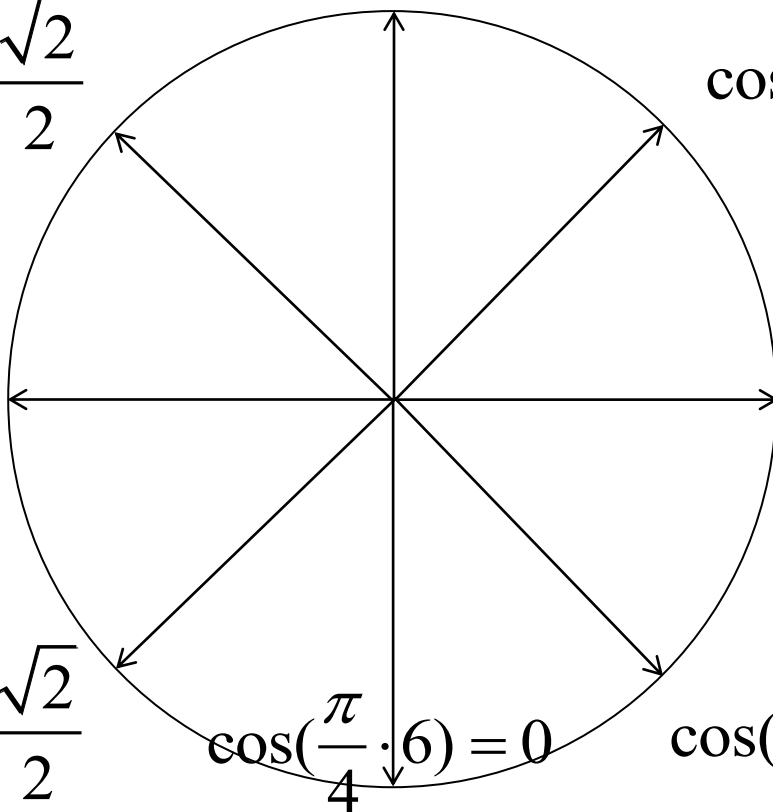
$$\cos\left(\frac{\pi}{4} \cdot 2\right) = 0$$

$$\cos\left(\frac{\pi}{4} \cdot 1\right) = \frac{\sqrt{2}}{2}$$

$$\cos\left(\frac{\pi}{4} \cdot 0\right) = 1$$

$$\cos\left(\frac{\pi}{4} \cdot 7\right) = \frac{\sqrt{2}}{2}$$

$$\cos\left(\frac{\pi}{4} \cdot 6\right) = 0$$



Normalised Frequency f

- We write the frequency in terms of the normalised frequency f as

$$0 \leq 2\pi f \leq \pi$$

- We can say the range of frequency f is

$$|f| \leq \frac{1}{2}$$

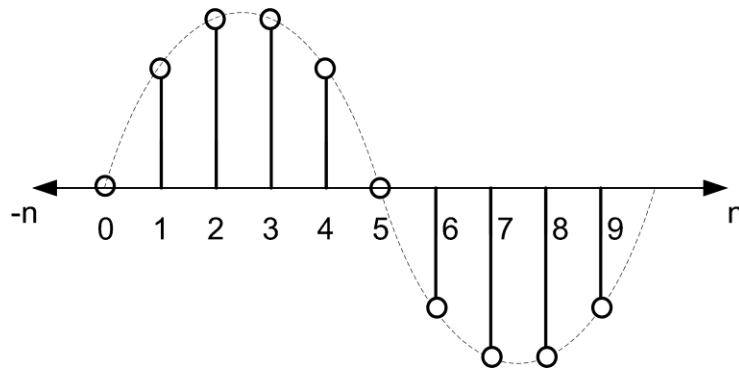
3 Sine signals with 2π radians apart are the same signals

- From $x_l(n) = \cos(\omega_l n + \theta)$
- let $\omega_l = \omega_0 + 2l\pi, \quad -\pi \leq \omega_0 \leq \pi$
- We have
$$\begin{aligned}x_l(n) &= \cos(\omega_l n + \theta) \\&= \cos((\omega_0 + 2\pi l)n + \theta) \\&= \cos(\omega_0 n + 2\pi l n + \theta) \\&= \cos(\omega_0 n + \theta) \\&= x(n)\end{aligned}$$

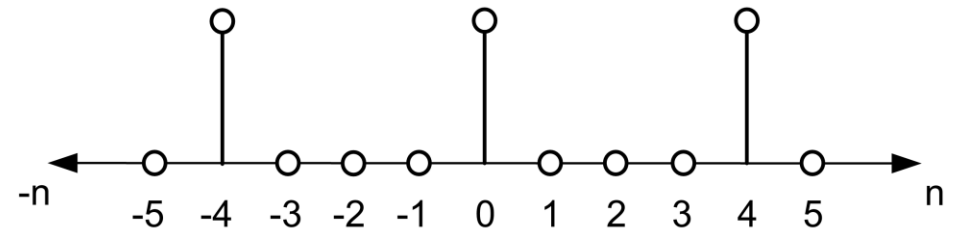
Discrete-Time Fourier Series

- Discrete-time periodic signals can be written in the forms of **Fourier Series**

Discrete-Time
Sinusoidal Signal



$\delta(n + 4l)$



Discrete-Time Fourier Series

- The Synthesis Equation

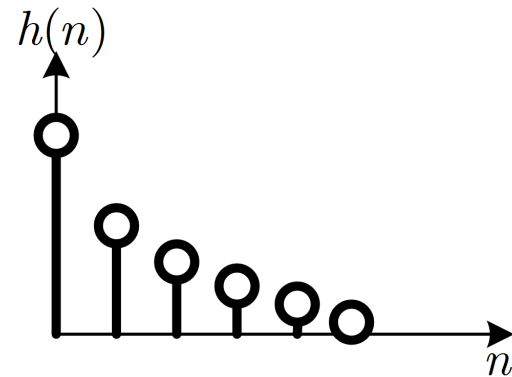
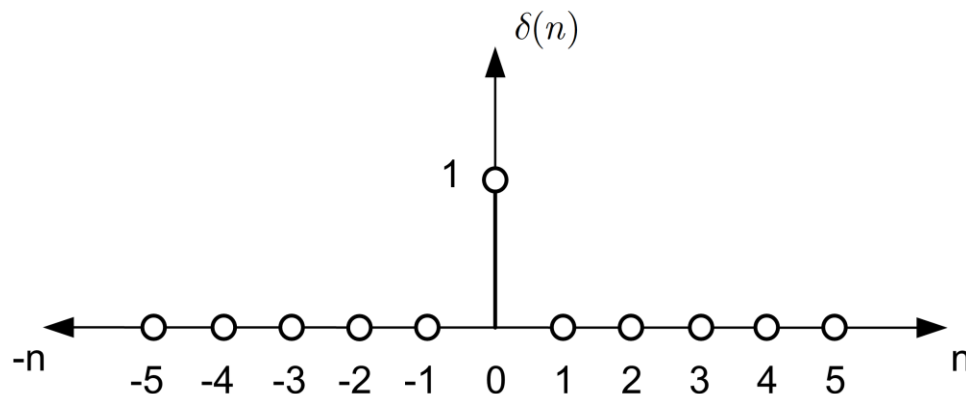
$$x(n) = \sum_{k=0}^{N-1} c_k e^{j\frac{2\pi kn}{N}}$$

- The Analysis I

$$c_k = \frac{1}{N} \sum_{n=0}^{N-1} x(n) e^{-j\frac{2\pi kn}{N}}$$

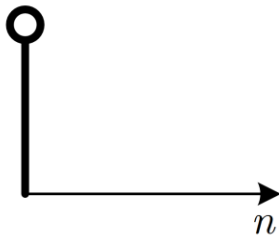
Discrete-Time Fourier Transform

- For **Aperiodic** signals, we use modify the signals as if its period is infinity then use the **Discrete-time Fourier Transform (DTFT)**.

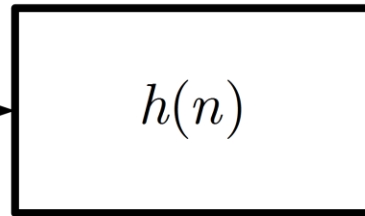


Time Domain

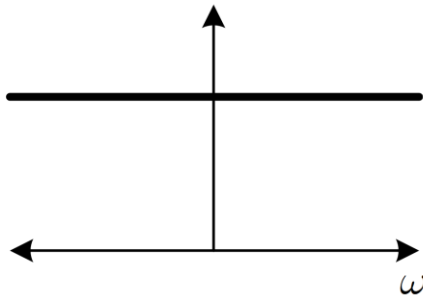
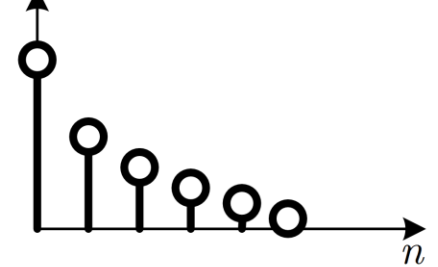
$\delta(n)$



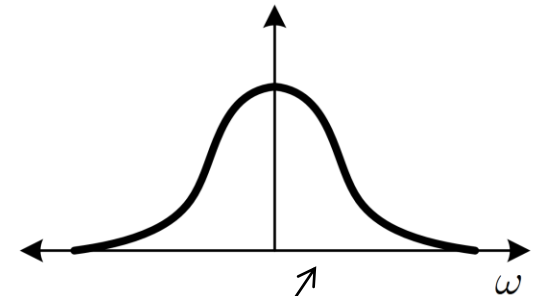
$h(n)$



$h(n)$

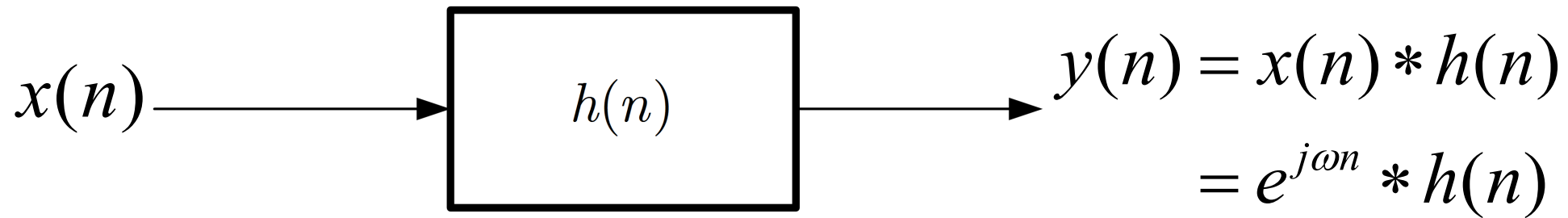


Frequency Domain



- Consider the system with input

$$x(n) = \cos \omega n + j \sin \omega n = e^{j\omega n}$$



The result of $h(n) * e^{j\omega n}$

$$\begin{aligned}y(n) &= \sum_{k=-\infty}^{\infty} h(k)x(n-k) \\&= \sum_{k=-\infty}^{\infty} h(k)e^{j\omega(n-k)} \\&= \sum_{k=-\infty}^{\infty} h(k)e^{-j\omega k} \cdot e^{j\omega n} \\&= \left(\sum_{k=-\infty}^{\infty} h(k)e^{-j\omega k} \right) e^{j\omega n} \\&= H(e^{j\omega})x(n)\end{aligned}$$

$$H(e^{j\omega}) = \sum_{k=-\infty}^{\infty} h(k)e^{-j\omega k}$$

The Discrete-Time Fourier Transform

- The DTFT of $h(n)$ is then

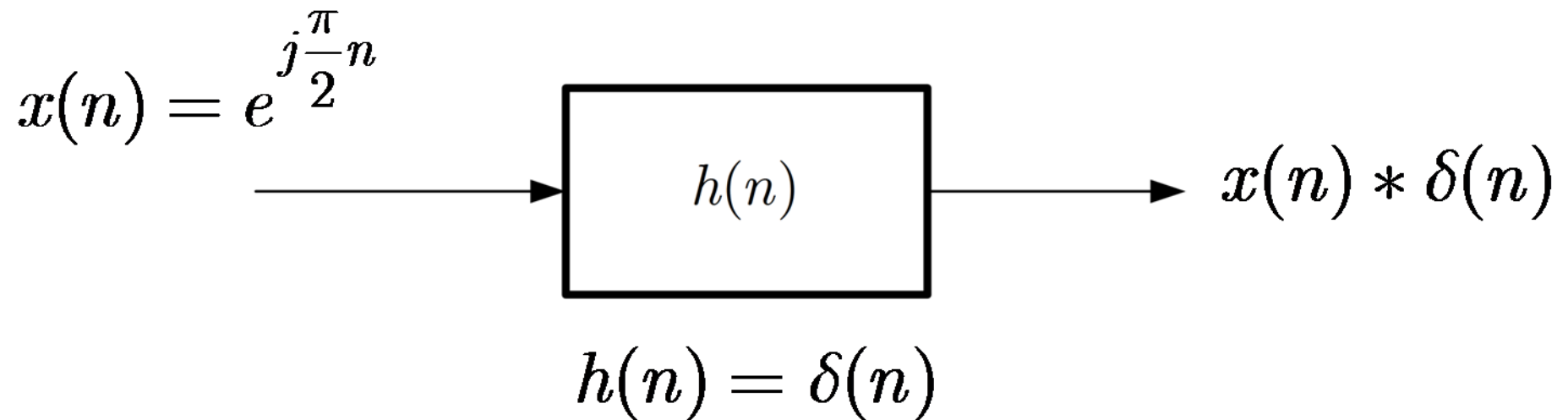
$$H(e^{j\omega}) \equiv \sum_{n=-\infty}^{\infty} h(n)e^{-j\omega n}$$

- Where ω is the digital frequency.

The delta function as the impulse response

- Let us determine the output of a system whose impulse response is described by the

$$h(n) = \delta(n)$$



The output of the system:

$$\begin{aligned}y(n) &= x(n) \\y(n) &= H(e^{j(\pi/2)}) \cdot x(n) \\&= \left(\sum_{k=-\infty}^{\infty} [\delta(k)] e^{-j(\pi/2)k} \right) \cdot e^{j(\pi/2)n} \\&= \left(\delta(0) e^{-j(\pi/2)0} \right) \cdot e^{j(\pi/2)n} \\&= (1) \cdot e^{j(\pi/2)n} \\&= e^{j(\pi/2)n} \\&= x(n)\end{aligned}$$

The delta function as the impulse response : DTFT

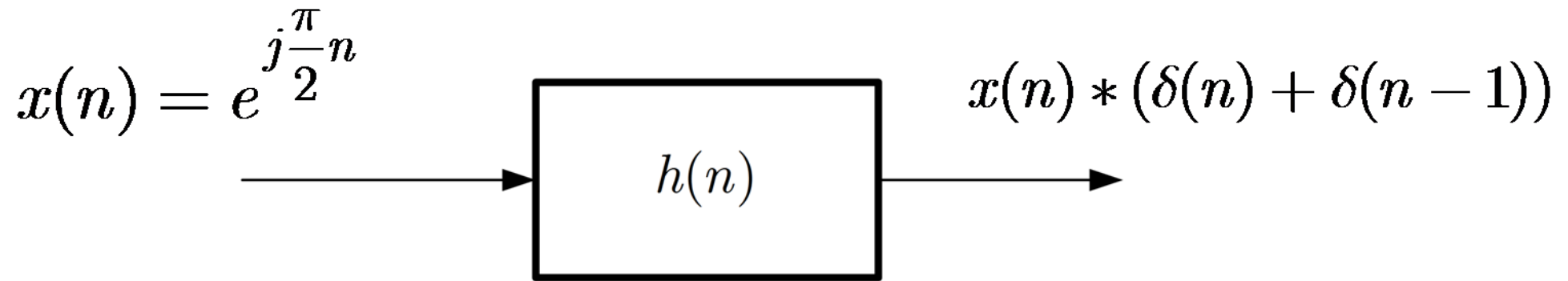
- By the notation of delta function, we have

$$H(e^{j\omega}) = \delta(0) \cdot e^{-j\frac{\pi}{2}(0)} = 1 \cdot 1 = 1$$

- So the DTFT of delta function is one.
- **This means the frequency response of the system is a constant function.**
- A highly preferable property of audio amplifiers.

The delta function and a unit delayed as the impulse response

$$h(n) = \delta(n) + \delta(n - 1)$$



$$h(n) = \delta(n) + \delta(n - 1)$$

The Discrete-Time Fourier Transform

$$\begin{aligned} H(e^{j\omega}) &= \sum_{k=-\infty}^{\infty} h(n)e^{-j\omega k} \\ &= \sum_{k=-\infty}^{\infty} [\delta(k) + \delta(k-1)]e^{-j\omega k} \end{aligned}$$

The output of the system

$$\begin{aligned}y(n) &= H(e^{j(\pi/2)}) \cdot x(n) \\&= \left(\sum_{k=-\infty}^{\infty} [\delta(k) + \delta(k-1)] e^{-j(\pi/2)k} \right) \cdot e^{j(\pi/2)n} \\&= \left(\delta(0) e^{-j(\pi/2)0} + \delta(1-1) e^{-j(\pi/2)} \right) \cdot e^{j(\pi/2)n} \\&= \left(1 + e^{-j(\pi/2)} \right) \cdot e^{j(\pi/2)n} \\&= e^{j(\pi/2)n} + e^{-j(\pi/2)} \cdot e^{j(\pi/2)n} \\&= x(n) + e^{-j(\pi/2)} \cdot x(n)\end{aligned}$$

The delta function and a unit delayed as the impulse response: DTFT

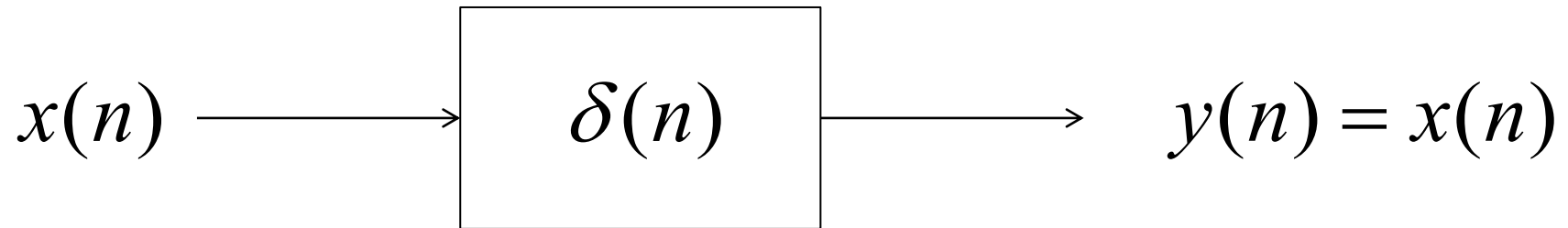
- In general, for any particular frequency

$$\begin{aligned} H(e^{j\omega}) &= \sum_{k=-\infty}^{\infty} [\delta(k) + \delta(k-1)] e^{-j\omega k} \\ &= 1 \cdot e^{-j\omega(0)} + 1 \cdot e^{-j\omega(1)} \\ &= 1 + e^{-j\omega} \end{aligned}$$

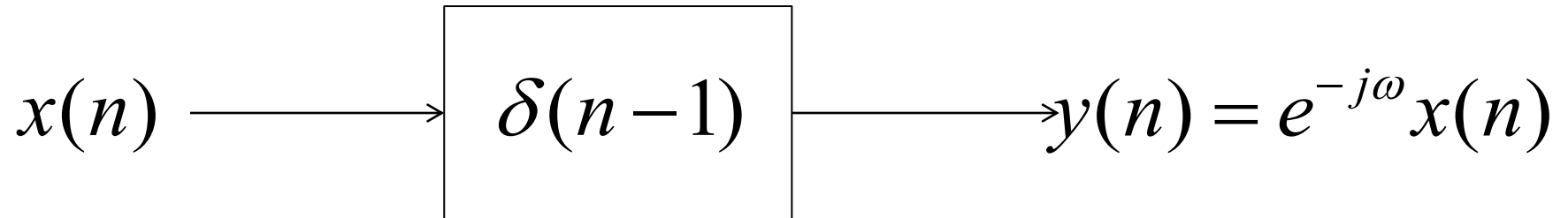
Output at a particular frequency for

$$h(n) = \delta(n) \quad h(n) = \delta(n-1)$$

- For a delta function as impulse response



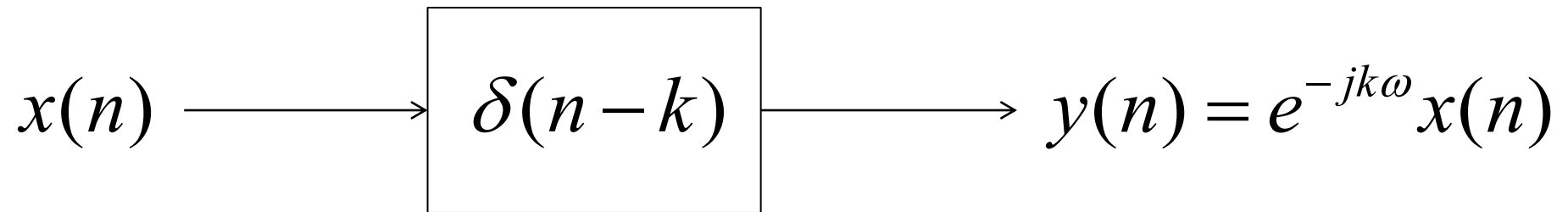
- For a unit delayed delta function as impulse response



Output at a particular frequency for

$$h(n) = \delta(n - k)$$

- For any k delayed delta function



Geometric Series and DTFT

- **Infinite geometric series:**

$$\sum_{n=0}^{\infty} a^n = \frac{1}{1-a}, \quad a < 1$$

- **Finite geometric series:**

$$\sum_{n=0}^{L-1} a^n = \begin{cases} \frac{1-a^L}{1-a}, & a \neq 1 \\ L, & a = 1 \end{cases}$$

Example 3.2: Determine the DTFT of

$$x(n) = (0.5)^n u(n)$$

- Solution:

$$\text{From } X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n}$$

Then

$$\begin{aligned} X(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} (0.5)^n u(n) e^{-j\omega n} = \sum_{n=0}^{\infty} (0.5)^n e^{-j\omega n} \\ &= \sum_{n=0}^{\infty} (0.5e^{-j\omega})^n \\ &= \frac{1}{1 - 0.5e^{-j\omega}} = \frac{e^{j\omega}}{e^{j\omega} - 0.5} \end{aligned}$$

$$\sum_{n=0}^{\infty} a^n = \frac{1}{1-a}, \quad a < 1$$

Example 3.3: Determine the DTFT of

$$x(n) = \begin{cases} 0.5, & 0 \leq n \leq L-1 \\ 0, & \textit{otherwise} \end{cases}$$

Solution:

$$X(e^{j\omega}) = \sum_{n=0}^{L-1} (0.5)e^{-j\omega n}$$

$$= 0.5 \sum_{n=0}^{L-1} (e^{-j\omega})^n$$

$$= 0.5 \frac{1 - e^{-j\omega L}}{1 - e^{-j\omega}}$$

$$\sum_{n=0}^{L-1} a^n = \begin{cases} L, & a = 1 \\ \frac{1 - a^L}{1 - a}, & a \neq 1 \end{cases}$$

Example 3.4: DTFT in general

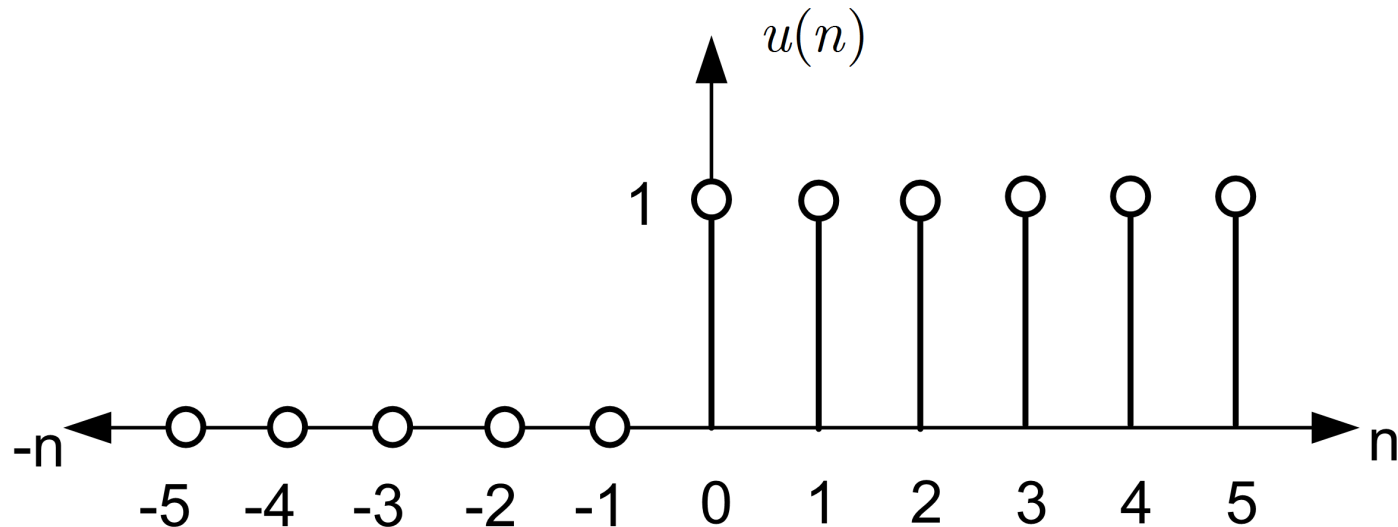
- Determine the DTFT of $x(n) = \{1, \underset{\uparrow}{2}, 3, 4, 5\}$
- Solution:

$$\begin{aligned} X(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n} \\ &= e^{j\omega} + 2 + 3e^{-j\omega} + 4e^{-j2\omega} + 5e^{-j3\omega} \end{aligned}$$

Note the position of up-arrow symbol $\uparrow$

For Sequences with the Unit Step function

- From
$$u(n) = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$



Example 3.5: Determine the DTFT of

$$x(n) = u(n) - u(n - 6)$$

- Solution: From **the Finite geometric series**:

$$\begin{aligned} X(e^{j\omega}) &= \sum_{n=0}^5 e^{-j\omega n} \\ &= \frac{1 - e^{-j6\omega}}{1 - e^{-j\omega}} \end{aligned}$$

$$\sum_{n=0}^{L-1} a^n = \begin{cases} L, & a = 1 \\ \frac{1 - a^L}{1 - a}, & a \neq 1 \end{cases}$$

Example 3.6: Determine the DTFT of

$$x(n) = u(-n)$$

- Solution:

Change the
direction
of index
 $m = -n$

$$\begin{aligned} X(e^{j\omega}) &= \sum_{n=-\infty}^0 e^{-j\omega n} \\ &= \sum_{m=0}^{\infty} e^{j\omega m} \\ &= \sum_{n=0}^{\infty} (e^{j\omega})^n \\ &= \frac{1}{1 - e^{j\omega}} \end{aligned}$$

$$\boxed{\sum_{n=0}^{\infty} a^n = \frac{1}{1-a}, \quad a < 1}$$

Example 3.7: Determine the DTFT of

$$x(n) = 2^n u(-n)$$

- Solution:

$$X(e^{j\omega}) = \sum_{n=-\infty}^0 2^n e^{-j\omega n}$$

$$= \sum_{m=0}^{\infty} 2^{-m} e^{j\omega m}$$

$$= \sum_{m=0}^{\infty} \frac{e^{j\omega m}}{2^m}$$

$$= \sum_{n=0}^{\infty} \left(\frac{e^{j\omega}}{2}\right)^n$$

$$= \frac{1}{1 - e^{j\omega}/2}$$

$$= \frac{2}{2 - e^{j\omega}}$$

$$\text{DTFT}(\mathbf{x} * \mathbf{h}) = \text{DTFT}(\mathbf{x}) \text{DTFT}(\mathbf{h})$$

- We usually compute the convolution in $x(n)$ and $h(n)$ in the time domain.
- But if $x(n)$ and $h(n)$ each has DTFT $X(e^{j\omega})$, $H(e^{j\omega})$
- then

$$\begin{aligned} F\{x(n) * h(n)\} &= F\{x(n)\} F\{h(n)\} \\ &= X(e^{j\omega}) H(e^{j\omega}) \end{aligned}$$

Example 3.8: Find the result of $x(n)*h(n)$ using DTFT

$$x(n) = h(n) = \{1, \underset{\uparrow}{1}, 1\}$$

- Solution:

$$\begin{aligned} X(e^{j\omega}) &= H(e^{j\omega}) = \sum_{n=-1}^1 x(n)e^{-j\omega n} \\ &= 1 \cdot e^{-j\omega(-1)} + 1 \cdot e^{-j\omega(0)} + 1 \cdot e^{-j\omega(1)} \\ &= e^{j\omega} + 1 + e^{-j\omega} \end{aligned}$$

- So we say that the output of the system

$$\begin{aligned} Y(e^{j\omega}) &= X(e^{j\omega})H(e^{j\omega}) \\ &= (e^{j\omega} + 1 + e^{-j\omega})^2 \\ &= e^{j2\omega} + 2e^{j\omega} + 3 + 2e^{-j\omega} + e^{-j2\omega} \end{aligned}$$

- Or we can write

$$Y(e^{j\omega}) = 1 \cdot e^{j2\omega} + 2 \cdot e^{j\omega} + 3 \cdot e^{j\omega} + 2 \cdot e^{-j\omega} + 1 \cdot e^{-j2\omega}$$

$y(n)$ are the coefficients of the DTFT result

- If we compare

$$Y(e^{j\omega}) = 1 \cdot e^{j2\omega} + 2 \cdot e^{j\omega} + 3 \cdot e^{j\omega} + 2 \cdot e^{-j\omega} + 1 \cdot e^{-j2\omega}$$

- With

$$Y(e^{j\omega}) = \sum_{n=-2}^2 y(n) e^{-j\omega n}$$

- This means that we can use the coefficients for $y(n)$ as

$$y(n) = \{1, 2, \underset{\uparrow}{3}, 2, 1\}$$

Homework 3.1

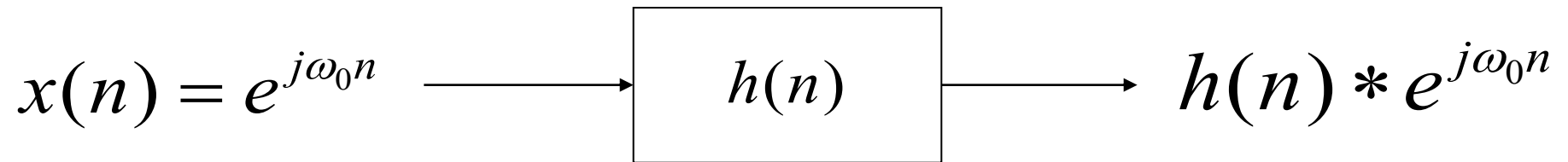
Consider the system whose impulse response is

$$h(n) = [\underset{\uparrow}{1}, 1, 1]$$

And with input $x(n) = [1, \underset{\uparrow}{2}, 3]$, determine the output of the system by means of

$$\begin{aligned} F\{h(n) * x(n)\} &= F\{h(n)\}F\{x(n)\} \\ &= H(e^{j\omega})X(e^{j\omega}) \end{aligned}$$

Frequency Responses (DTFT) of Systems at a particular frequency



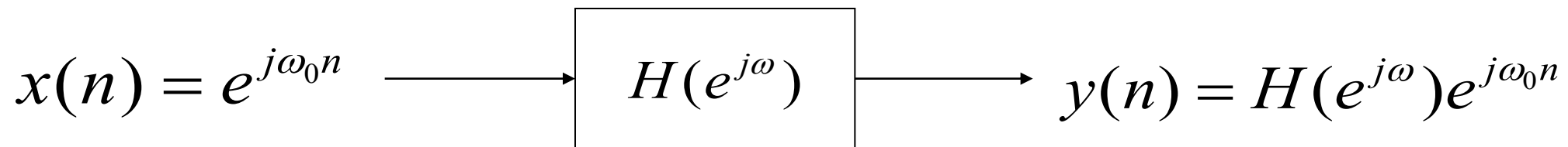
$$y(n) = h(n) * e^{j\omega_0 n} = \sum_{k=-\infty}^{\infty} h(k) e^{j\omega_0 (n-k)}$$

$$= \left[\sum_{k=-\infty}^{\infty} h(k) e^{-j\omega_0 k} \right] e^{j\omega_0 n}$$

$$= F(h(n)) \Big|_{\omega=\omega_0} e^{j\omega_0 n}$$



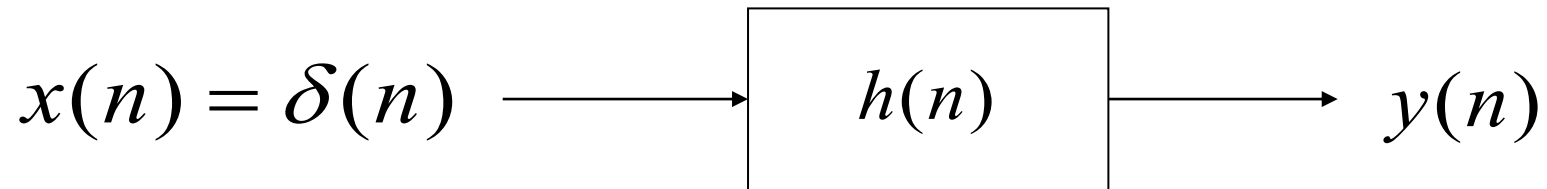
$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h(n)e^{-j\omega n}$$



$$Y(e^{j\omega}) = H(e^{j\omega})X(e^{j\omega})$$

Example 3.9:

Determine $X(e^{j\omega})$, $Y(e^{j\omega})$, $H(e^{j\omega})$ from the following system



$$h(n) = \{ \underset{\uparrow}{1}, 0.8, 0.2 \}$$

Solution 3.9

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n} = \sum_{n=-\infty}^{\infty} \delta(n)e^{-j\omega n} = 1e^{-j\omega 0} = 1$$

$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h(n)e^{-j\omega n} = 1e^{-j\omega 0} + 0.8e^{-j\omega} + 0.2e^{-j2\omega}$$

$$Y(e^{j\omega}) = H(e^{j\omega})X(e^{j\omega}) = 1 + 0.8e^{-j\omega} + 0.2e^{-j2\omega}$$

Poles and Zeros of the Frequency Response

- For a first-order system with impulse response $h(n)$, the DTFT (or Frequency Response) is

$$\begin{aligned} H(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} h(n)e^{-j\omega n} \\ &= \frac{e^{j\omega} - z_1}{e^{j\omega} - p_1} \end{aligned}$$

Poles and Zeros of the Frequency Response

- For a system with

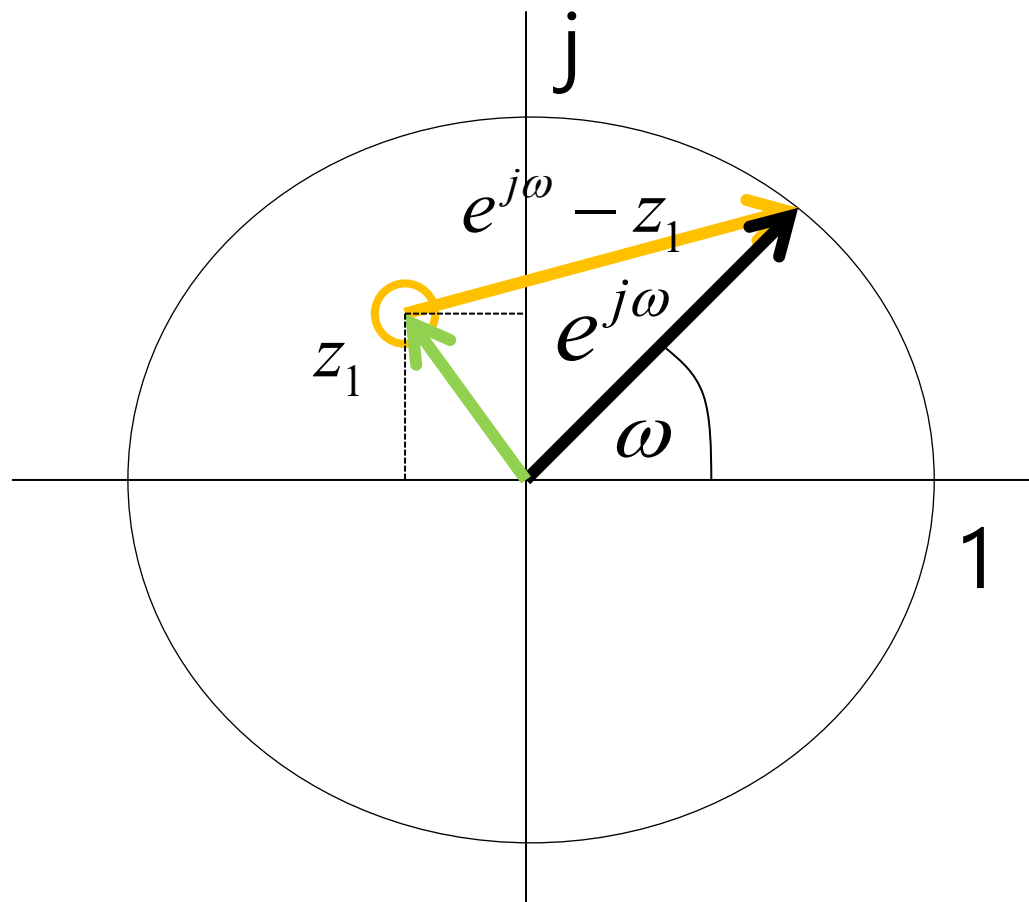
$$h(n) = \{\underset{\uparrow}{1}, -h(1)\}$$

- The DTFT (or Frequency Response) is

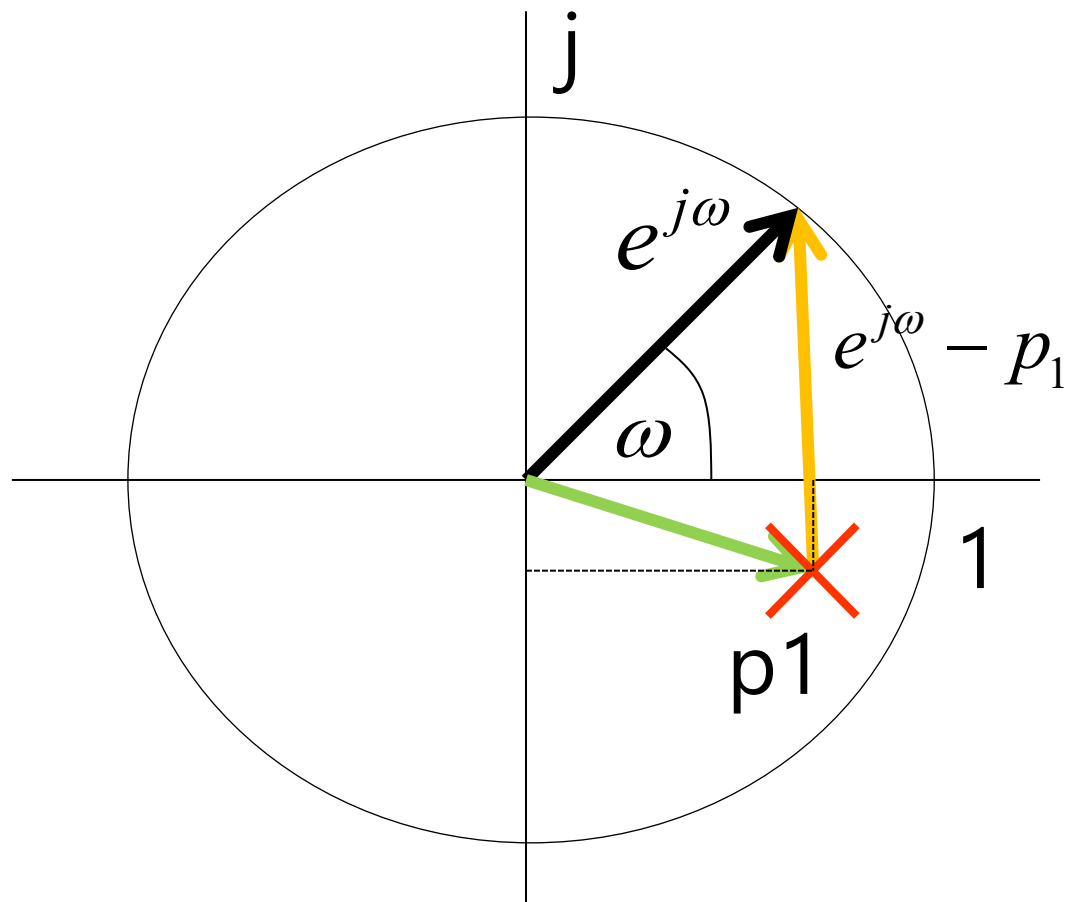
$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h(n)e^{-j\omega n} = 1 - h(1)e^{-j\omega}$$

$$H(e^{j\omega}) = e^{-j\omega}(e^{j\omega} - h(1)) = \frac{e^{j\omega} - h(1)}{e^{j\omega} - 0} = \frac{e^{j\omega} - z_1}{e^{j\omega} - p_1}$$

$$e^{j\omega} - z_1$$



$$e^{j\omega} - p_1$$



Example 3.10: Poles and Zeros of the Frequency Response

- For a system with

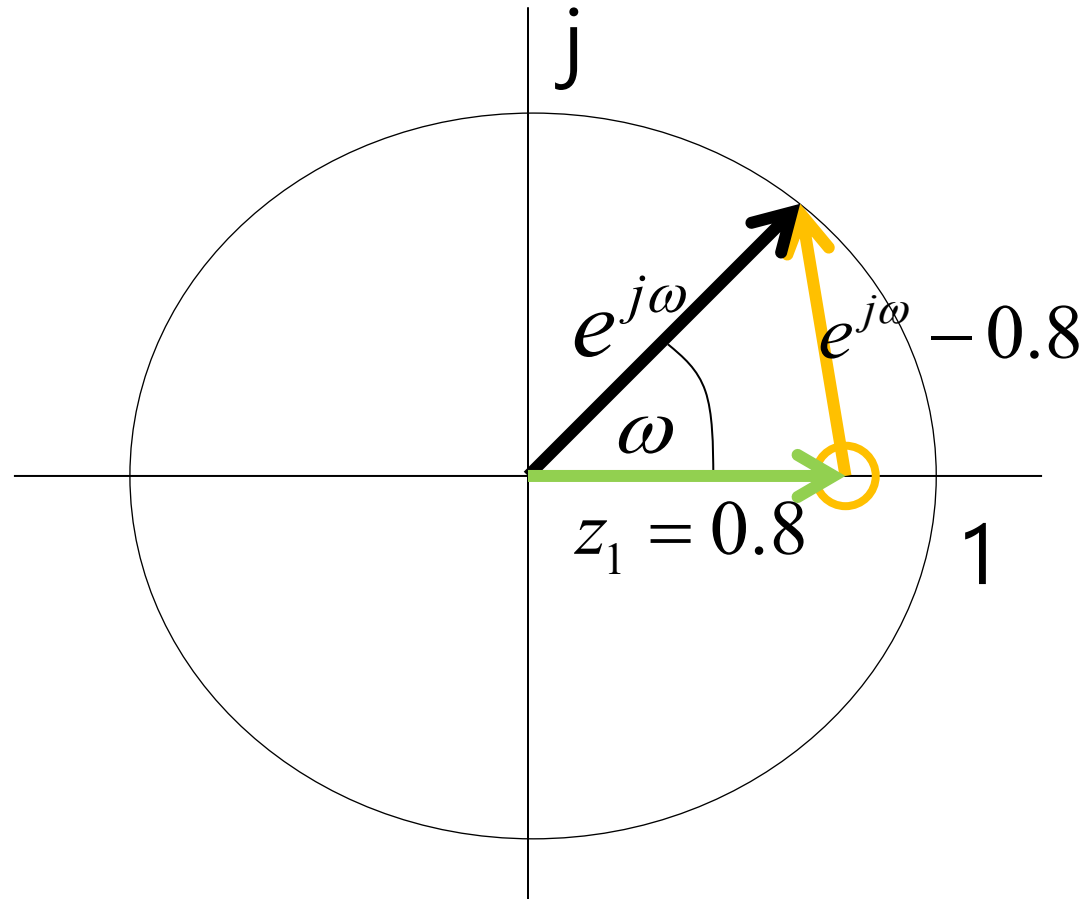
$$h(n) = \{\underset{\uparrow}{1}, -0.8\}$$

- The DTFT (or Frequency Response) is

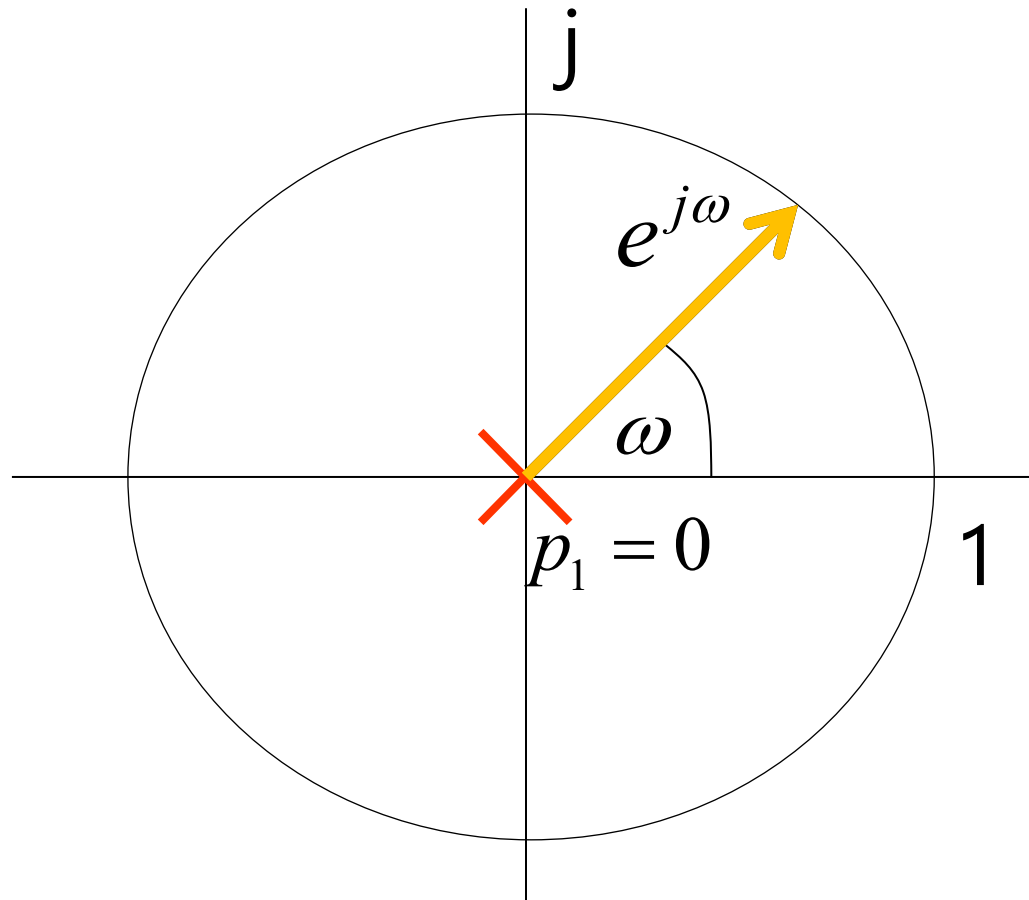
$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h(n)e^{-j\omega n} = 1 - 0.8e^{-j\omega}$$

$$H(e^{j\omega}) = e^{-j\omega}(e^{j\omega} - 0.8) = \frac{e^{j\omega} - 0.8}{e^{j\omega} - 0} = \frac{e^{j\omega} - z_1}{e^{j\omega} - p_1}$$

$$Z_1 = 0.8$$



$$p_1=0$$



Example 3.11: Poles and Zeros of the Frequency Response

- For a system with

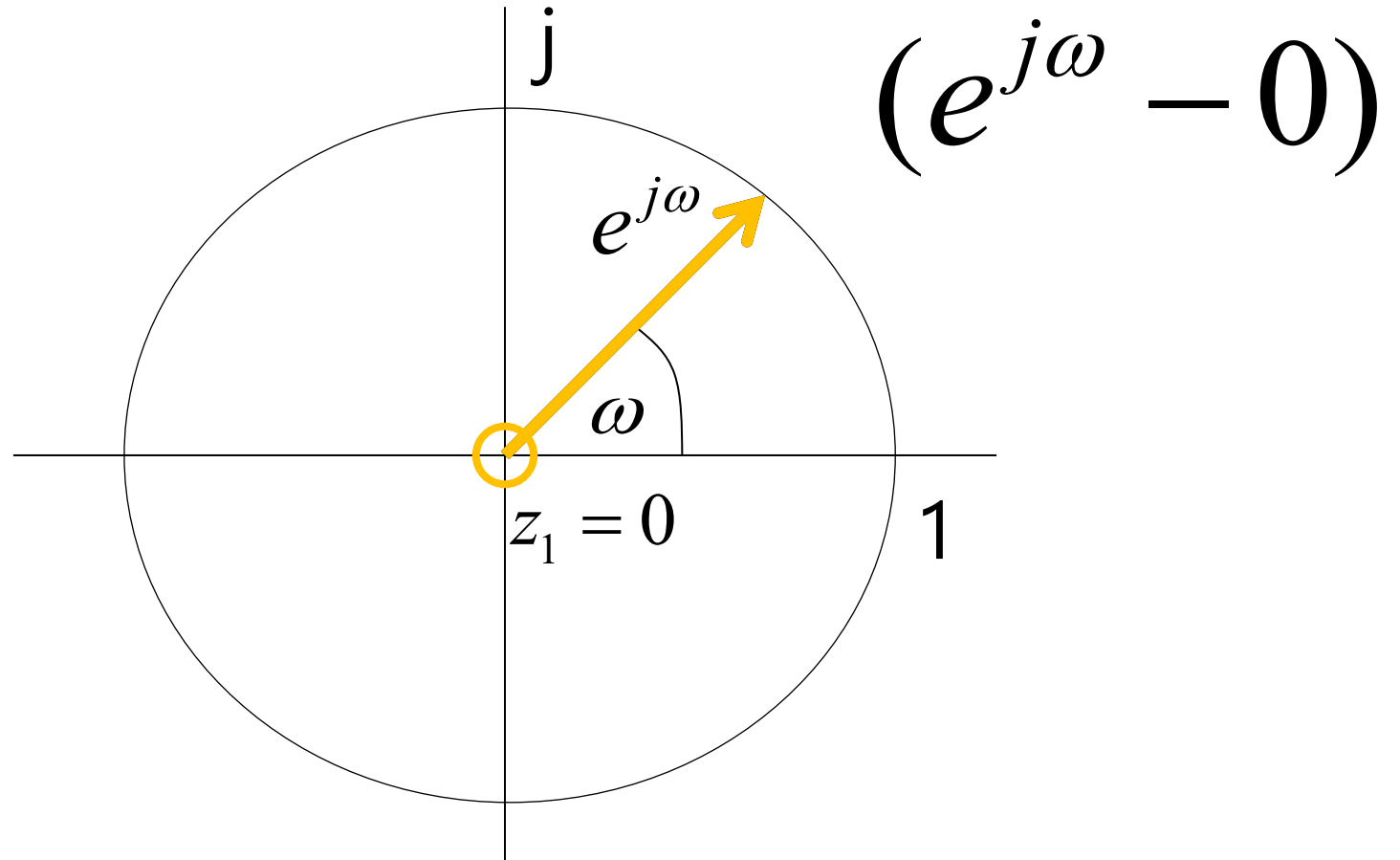
$$y(n) - 0.8y(n-1) = x(n)$$

- The DTFT of this system is

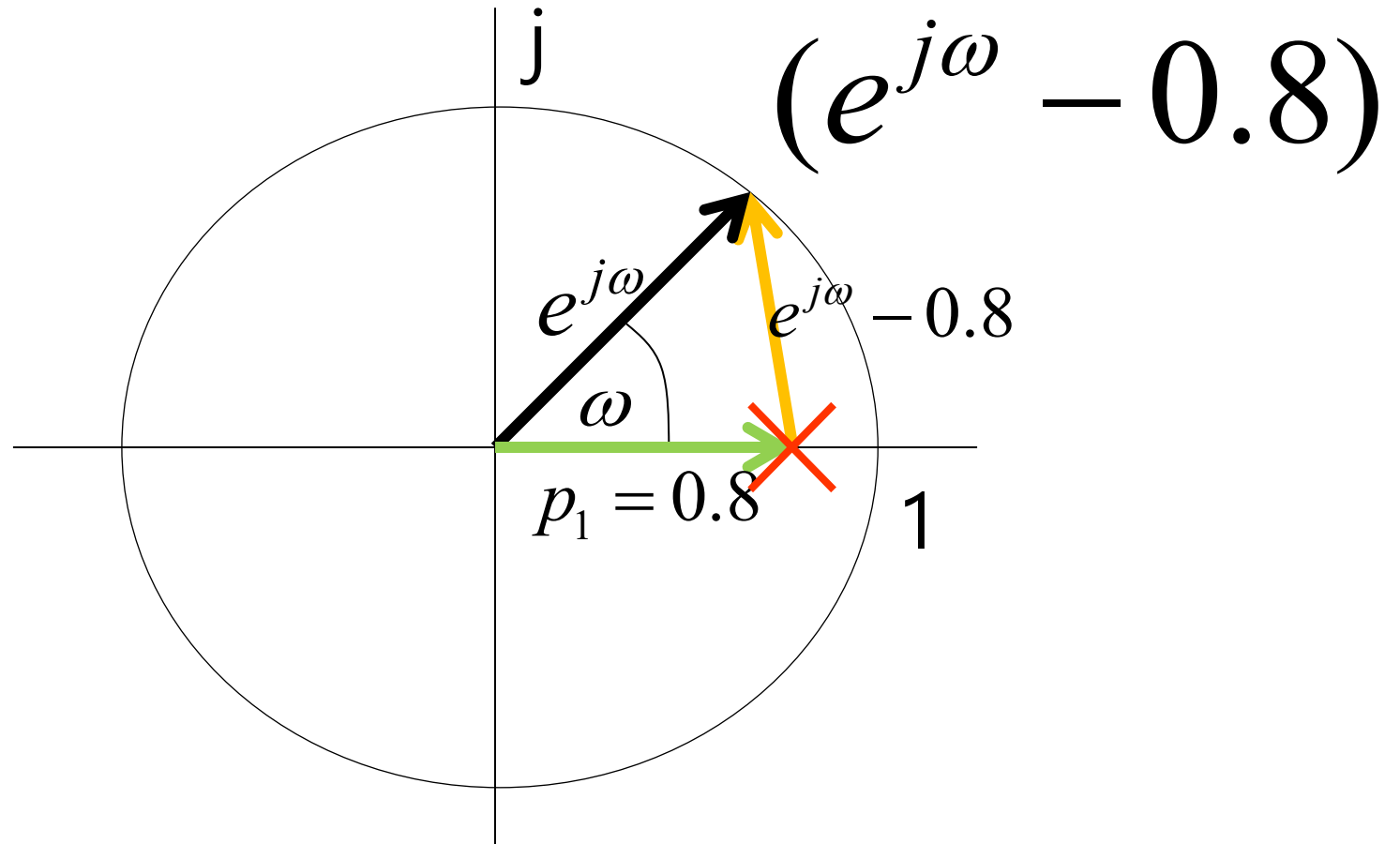
$$Y(e^{j\omega}) - 0.8e^{-j\omega}Y(e^{j\omega}) = X(e^{j\omega})$$

$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{1}{1 - 0.8e^{-j\omega}} = \frac{e^{j\omega}}{e^{j\omega} - 0.8} = \frac{(e^{j\omega} - z_1)}{(e^{j\omega} - p_1)}$$

$$z_1=0$$



$$p_1 = 0.8$$



Example 3.12: Poles and Zeros of the Frequency Response

- For a system with

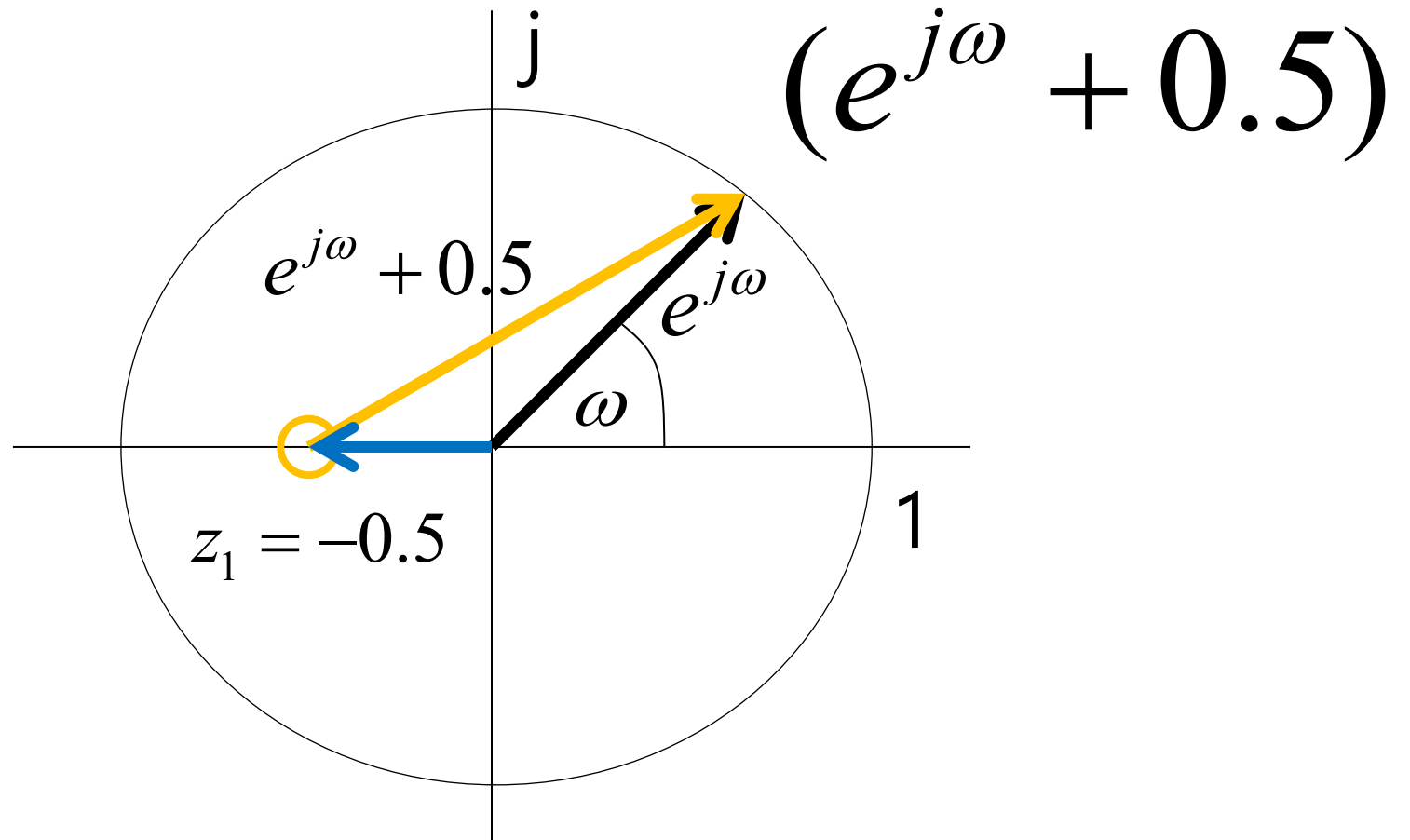
$$y(n) - 0.8y(n-1) = x(n) + 0.5x(n-1)$$

- The DTFT of this system is

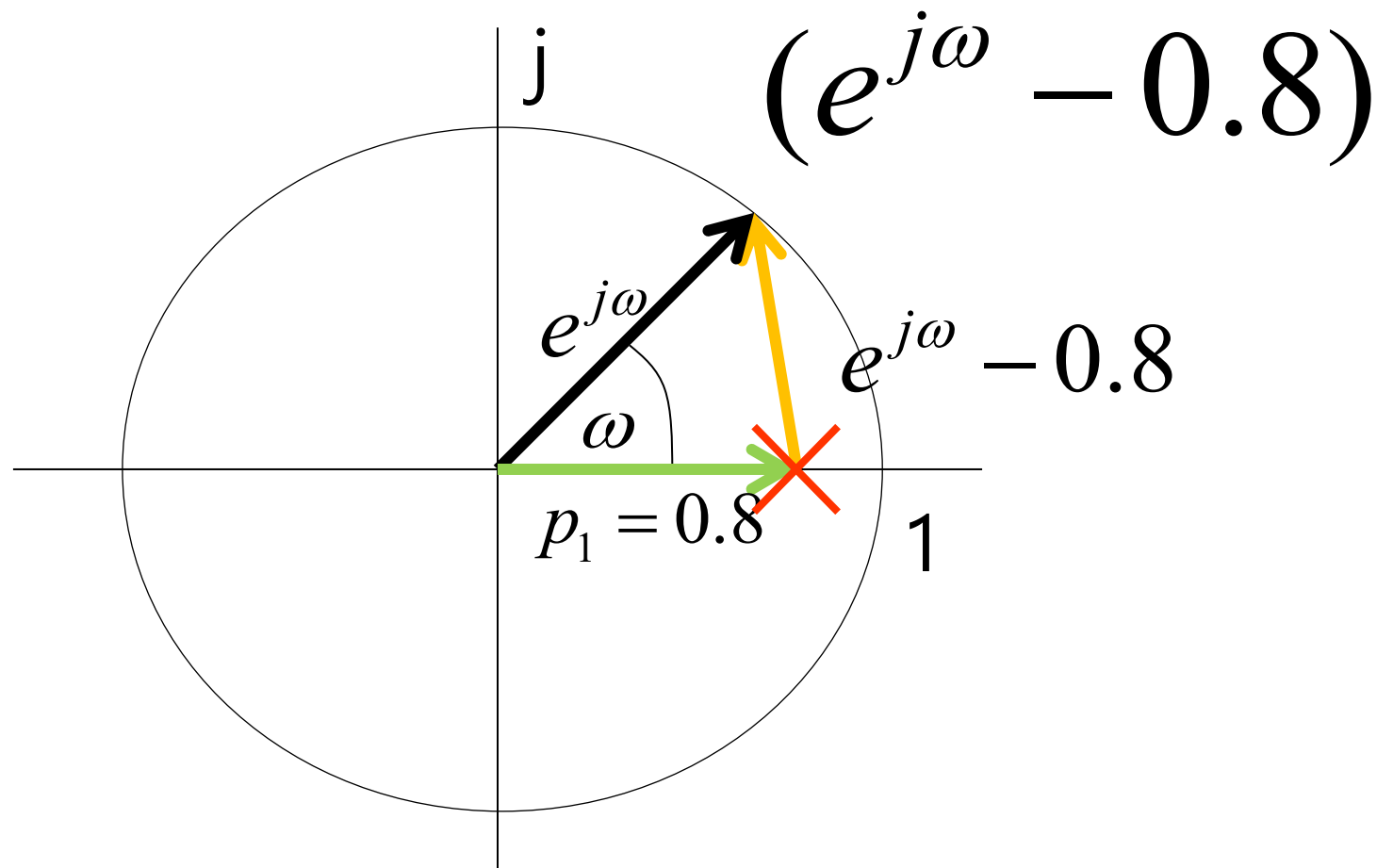
$$Y(e^{j\omega}) - 0.8e^{-j\omega}Y(e^{j\omega}) = X(e^{j\omega}) + 0.5e^{-j\omega}X(e^{j\omega})$$

$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{1 + 0.5e^{-j\omega}}{1 - 0.8e^{-j\omega}} = \frac{e^{j\omega} - (-0.5)}{e^{j\omega} - 0.8} = \frac{(e^{j\omega} - z_1)}{(e^{j\omega} - p_1)}$$

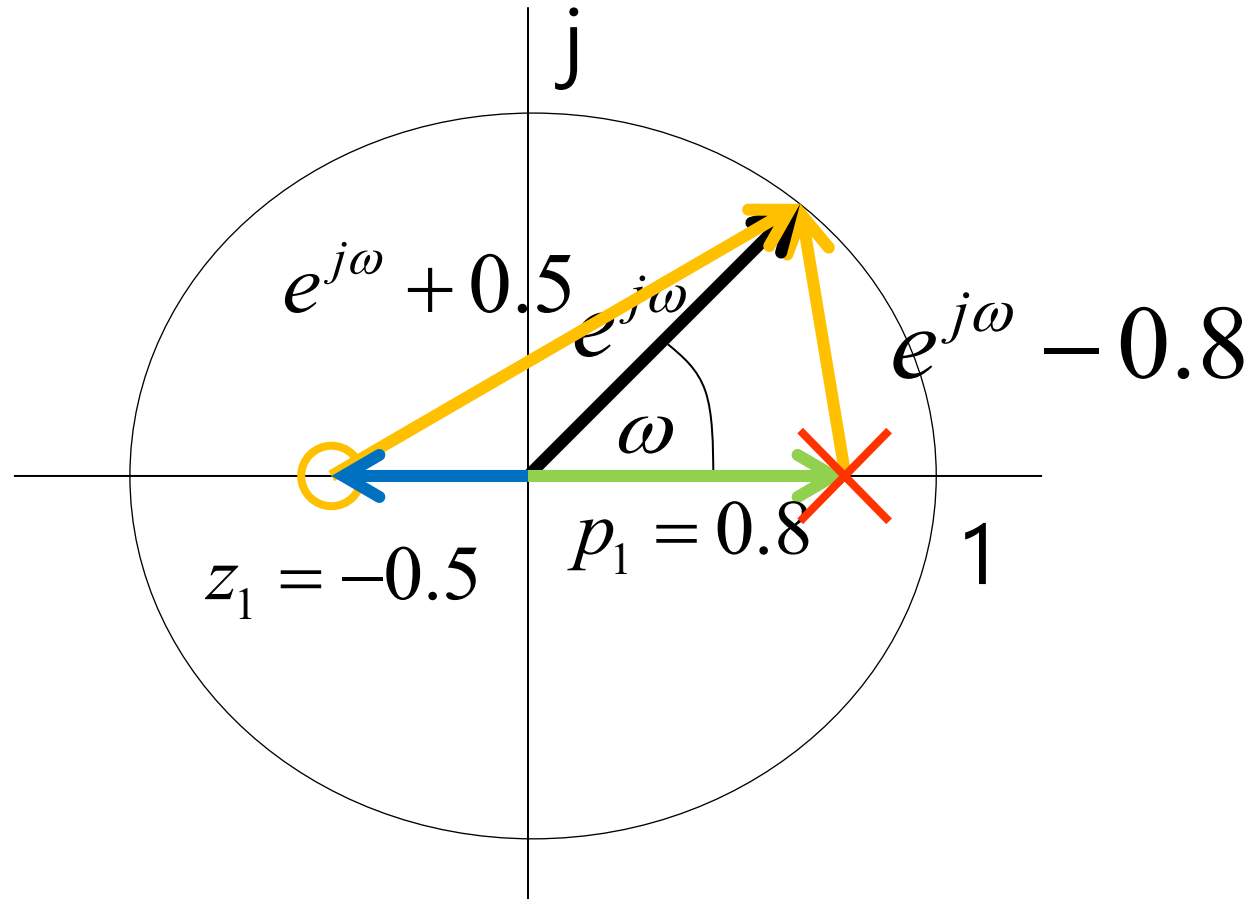
$$z_1 = -0.5$$



$$p_1 = 0.8$$



$$(e^{j\omega} + 0.5) / (e^{j\omega} - 0.8)$$



Poles and Zeros in General Form

$$\begin{aligned} H(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} h(n)e^{-j\omega n} = \frac{\sum_{k=0}^M b_k e^{-j\omega k}}{1 + \sum_{k=1}^N a_k e^{-j\omega k}} \\ &= b_0 \frac{(e^{j\omega} - z_1)(e^{j\omega} - z_2) \cdots (e^{j\omega} - z_M)}{(e^{j\omega} - p_1)(e^{j\omega} - p_2) \cdots (e^{j\omega} - p_N)} \end{aligned}$$

The Magnitude of Frequency Response

- At a particular frequency, the magnitude is

$$\begin{aligned} |H(e^{j\omega})| &= \left| b_0 \frac{(e^{j\omega} - z_1)(e^{j\omega} - z_2) \dots (e^{j\omega} - z_M)}{(e^{j\omega} - p_1)(e^{j\omega} - p_2) \dots (e^{j\omega} - p_N)} \right| \\ &= |b_0| \frac{|(e^{j\omega} - z_1)| |(e^{j\omega} - z_2)| \dots |(e^{j\omega} - z_M)|}{|(e^{j\omega} - p_1)| |(e^{j\omega} - p_2)| \dots |(e^{j\omega} - p_N)|} \end{aligned}$$

The Phase of Frequency Response

- At a particular frequency, the phase is

$$\angle H(e^{j\omega}) = \angle \text{numerator of } H(e^{j\omega}) \\ - \angle \text{denominator of } H(e^{j\omega})$$

$$\angle H(e^{j\omega}) = \tan^{-1} \left(\frac{\text{Im}\{H(e^{j\omega})\}}{\text{Re}\{H(e^{j\omega})\}} \right)$$

Example 3.13: System with a zero at 0.8

- From
$$h(n) = \{ \underset{\uparrow}{1}, -0.8 \}$$

- DTFT
$$H(e^{j\omega}) = \frac{e^{j\omega} - 0.8}{e^{j\omega} - 0}$$

- The magnitude is

$$|H(e^{j\omega})| = \frac{|e^{j\omega} - 0.8|}{|e^{j\omega} - 0|}$$

**At Frequencies
0 and $\pi/4$ radians/sample**

$$H(e^{j(0)}) = \frac{e^{j(0)} - 0.8}{e^{j(0)}} = 1 - 0.8 = 0.2 \angle 0$$

$$H(e^{j(\pi/4)}) = \frac{e^{j(\pi/4)} - 0.8}{e^{j(\pi/4)}} = 0.713 \angle 0.916$$

**At Frequencies
 $\pi/2$, $3\pi/4$ and π radians/sample**

$$H(e^{j(\pi/2)}) = \frac{e^{j(\pi/2)} - 0.8}{e^{j(\pi/2)}} = 1.28 \angle 0.674$$

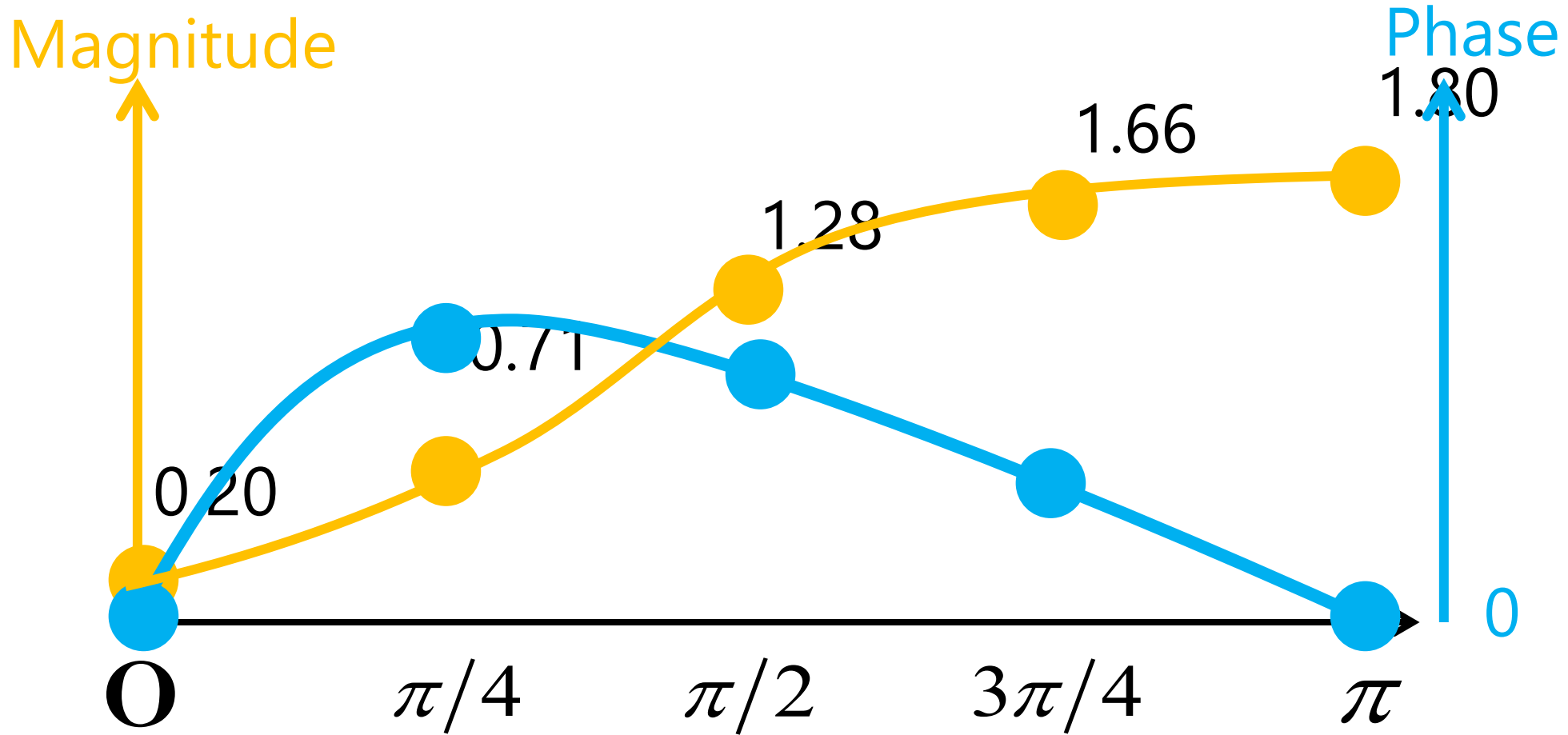
$$H(e^{j(3\pi/4)}) = \frac{e^{j(3\pi/4)} - 0.8}{e^{j(3\pi/4)}} = 1.66 \angle 0.346$$

$$H(e^{j(\pi)}) = \frac{e^{j(\pi)} - 0.8}{e^{j(\pi)}} = 1.8 \angle 0$$

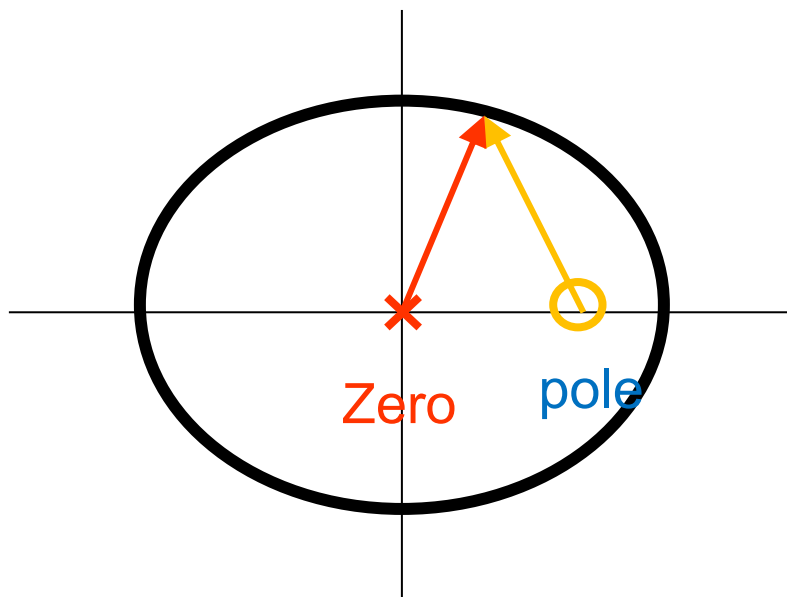
$$|H(e^{j\omega})| = \frac{|e^{j\omega} - 0.8|}{|e^{j\omega} - 0|}$$

0	0.2∠0
$\pi/4$	0.713∠0.916
$\pi/2$	1.28∠0.674
$3\pi/4$	1.66∠0.346
π	1.8∠0

Frequency Response: Magnitude and Phase



Frequency Response from Poles and Zeros



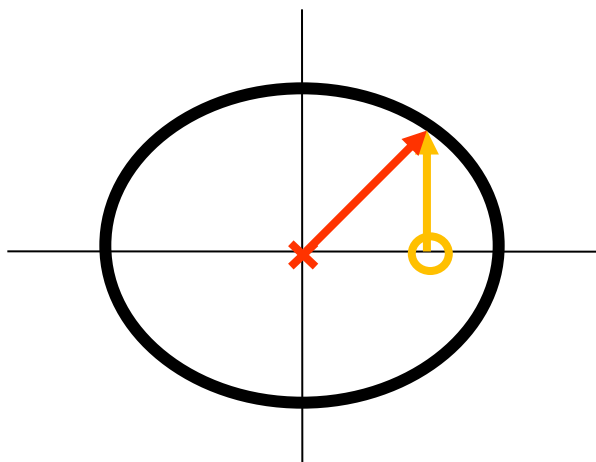
$$\left| H(e^{j\omega}) \right| = \frac{\left| e^{j\omega} - 0.8 \right|}{\left| e^{j\omega} - 0 \right|}$$

Example for Frequency Response

$$\omega = \pi/4$$

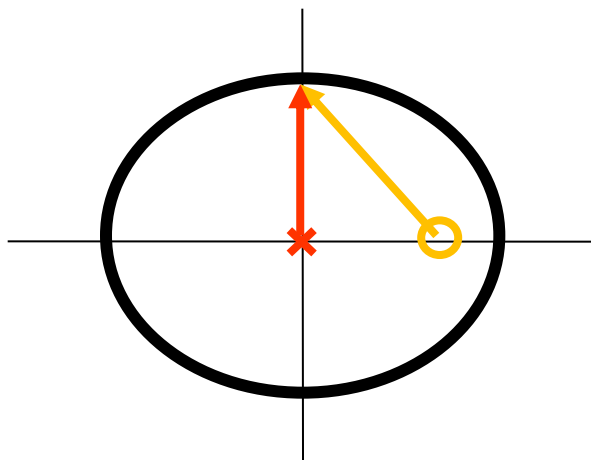
$$\omega = \pi/2$$

$$\omega = 3\pi/4$$



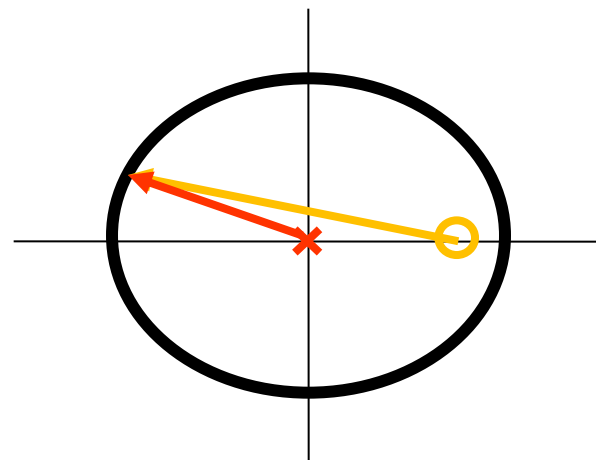
$$H(e^{j(\pi/4)})$$

$$= 0.713 \angle 0.916$$



$$H(e^{j(\pi/2)})$$

$$= 1.28 \angle 0.674$$



$$H(e^{j(3\pi/4)})$$

$$= 1.66 \angle 0.346$$

Example 3.14: System with a pole at 0.8

- From $y(n) - 0.8y(n-1) = x(n)$

- DTFT
$$H(e^{j\omega}) = \frac{e^{j\omega} - 0}{e^{j\omega} - 0.8}$$

- The magnitude is

$$|H(e^{j\omega})| = \frac{|e^{j\omega} - 0|}{|e^{j\omega} - 0.8|}$$

**At Frequencies
0 and $\pi/4$ radians/sample**

$$H(e^{j(0)}) = \frac{e^{j(0)}}{e^{j(0)} - 0.8} = \frac{1}{1 - 0.8} = 5 \angle 0$$

$$H(e^{j(\pi/4)}) = \frac{e^{j(\pi/4)}}{e^{j(\pi/4)} - 0.8} = 1.40 \angle -0.916$$

**At Frequencies
 $\pi/2$, $3\pi/4$ and π radians/sample**

$$H(e^{j(\pi/2)}) = \frac{e^{j(\pi/2)}}{e^{j(\pi/2)} - 0.8} = 0.78 \angle -0.674$$

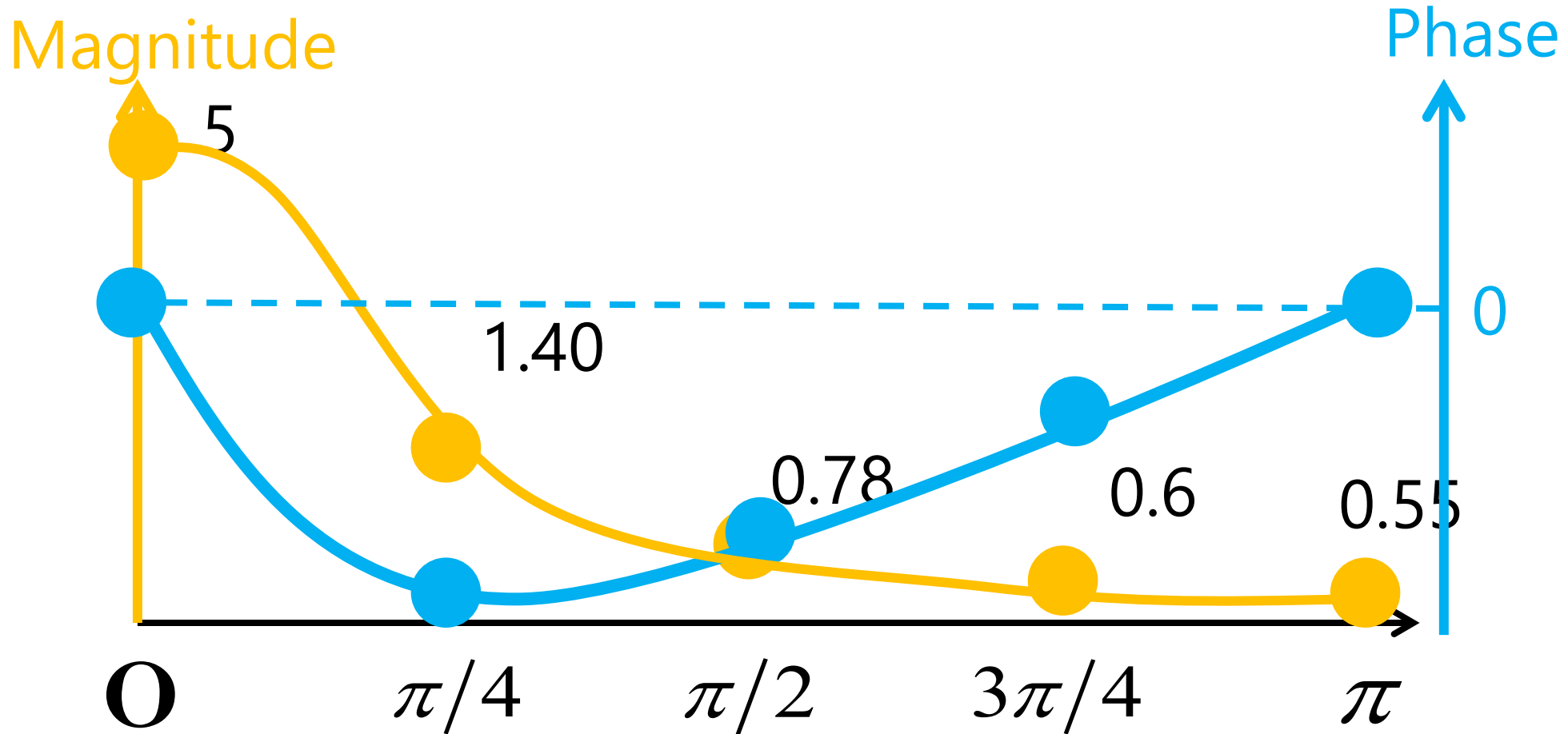
$$H(e^{j(3\pi/4)}) = \frac{e^{j(3\pi/4)}}{e^{j(3\pi/4)} - 0.8} = 0.6 \angle -0.346$$

$$H(e^{j(\pi)}) = \frac{e^{j(\pi)}}{e^{j(\pi)} - 0.8} = 0.55 \angle 0$$

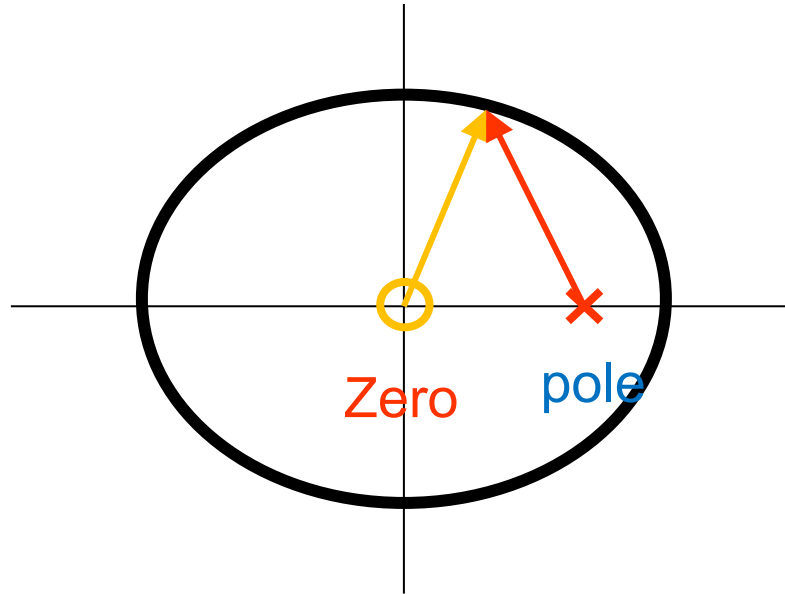
$$|H(e^{j\omega})| = \frac{|e^{j\omega} - 0.8|}{|e^{j\omega} - 0|}$$

0	5∠0
$\pi/4$	1.40∠−0.916
$\pi/2$	0.78∠−0.674
$3\pi/4$	0.6∠−0.346
π	0.55∠0

Frequency Response: Magnitude and Phase



Frequency Response from Poles and Zeros



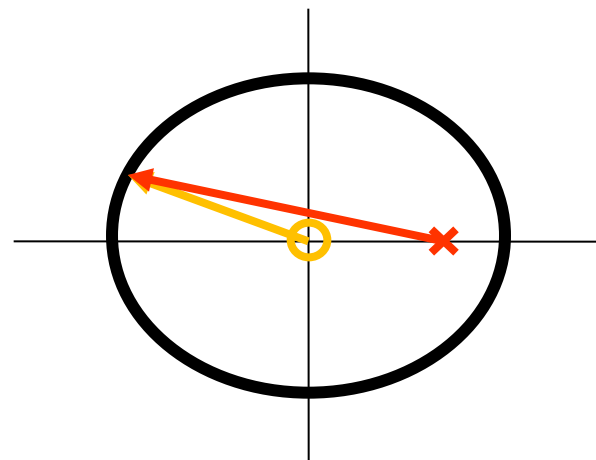
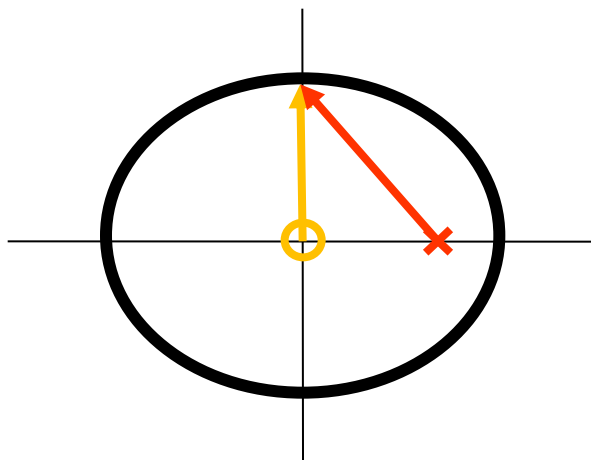
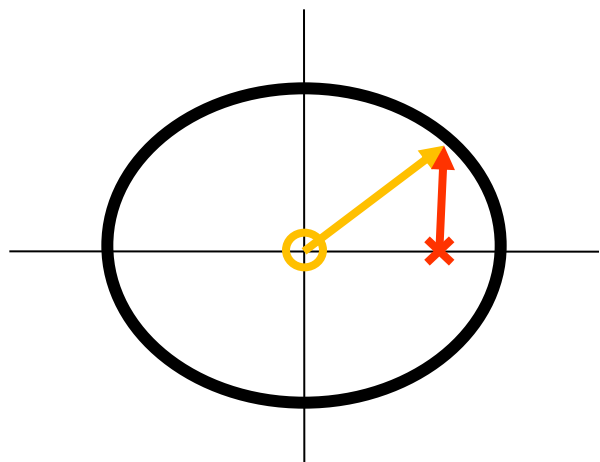
$$\left| H(e^{j\omega}) \right| = \frac{\left| e^{j\omega} \right|}{\left| e^{j\omega} - 0.8 \right|}$$

Example for Frequency Response

$$\omega = \pi/4$$

$$\omega = \pi/2$$

$$\omega = 3\pi/4$$



$$1.40 \angle -0.916$$

$$0.78 \angle -0.674$$

$$0.6 \angle -0.346$$

Finding $h(n)$ from DTFT of the single pole system

- From

$$H(e^{j\omega}) = \frac{e^{j\omega} - 0}{e^{j\omega} - 0.8} = \frac{1}{1 - 0.8e^{-j\omega}}$$

- Remember the **infinite geometric formula**

$$\sum_{n=0}^{\infty} a^n = \frac{1}{1-a}, \quad a < 1$$

- And the DTFT of $h(n)$

$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h(n)e^{-j\omega n}$$

$$H(e^{j\omega}) = \frac{1}{1 - 0.8e^{-j\omega}}$$

$$= \sum_{n=0}^{\infty} (0.8e^{-j\omega})^n$$

$$= \sum_{n=0}^{\infty} 0.8^n e^{-j\omega n}$$

Invoke the unit step
property

$$= \sum_{n=-\infty}^{\infty} \underbrace{0.8^n u(n)}_{=h(n)} e^{-j\omega n}$$

$$\therefore h(n) = 0.8^n u(n)$$

Example 3.15:

- Determine the output $y(n)$ of a system whose impulse response is

$$h(n) = \left(\frac{1}{2}\right)^n u(n)$$

- With input $x(n) = Ae^{j\frac{\pi n}{2}}$

Solution

- Notice that we are interested the response of the system at the particular frequency $\omega_0 = \frac{\pi}{2}$
- So
$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h(n)e^{-j\omega n} = \sum_{n=-\infty}^{\infty} 0.5^n u(n)e^{-j\omega n}$$
$$= \frac{1}{1 - 0.5e^{-j\omega}}$$

From $\sum_{n=0}^{\infty} a^n = \frac{1}{1-a}, \quad a < 1$
- DTFT
$$H(e^{j\frac{\pi}{2}}) = \frac{1}{1 - 0.5e^{-j\frac{\pi}{2}}} = \frac{1}{1 + j0.5} = \frac{2}{\sqrt{5}} e^{-j26.6}$$

The Output $y(n)$ at a Particular Frequency

- From

$$y(n) = H(e^{j\omega})x(n)$$

- Then

$$\begin{aligned} y(n) &= \left(\frac{2}{\sqrt{5}} e^{-j26.6} \right) A e^{j\frac{\pi n}{2}} \\ &= \frac{2}{\sqrt{5}} A e^{j\left(\frac{\pi n}{2} - 26.6\right)} \end{aligned}$$

$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h(n)e^{-j\omega n}$$

$$= \frac{1}{1 - \frac{1}{2}e^{-j\omega}}$$

$$= \frac{1}{1 - \frac{1}{2}e^{-j\omega}} \frac{e^{j\omega}}{e^{j\omega}} = \frac{e^{j\omega}}{e^{j\omega} - \frac{1}{2}}$$

Frequency Response from Difference Equations

- A difference equation is in the form of

$$y(n) + \sum_{l=1}^N a_l y(n-l) = \sum_{m=0}^M b_m x(n-m)$$

- With $x(n) = e^{j\omega n}$, $y(n) = H(e^{j\omega})e^{j\omega n}$ and

$$H(e^{j\omega})e^{j\omega n} + \sum_{l=1}^N a_l H(e^{j\omega})e^{j\omega(n-l)} = \sum_{m=0}^M b_m e^{j\omega(n-m)}$$

The Frequency Response (DTFT)

$$H(e^{j\omega})e^{j\omega n} + \sum_{l=1}^N a_l H(e^{j\omega})e^{j\omega(n-l)} = \sum_{m=0}^M b_m e^{j\omega(n-m)}$$

$$\left(1 + \sum_{l=1}^N a_l H(e^{j\omega})e^{j\omega(-l)}\right) H(e^{j\omega})e^{j\omega n} = \left(\sum_{m=0}^M b_m e^{j\omega(-m)}\right) e^{j\omega n}$$

- Then the DTFT is
$$H(e^{j\omega}) = \frac{\sum_{m=0}^M b_m e^{-j\omega m}}{1 + \sum_{l=1}^N a_l e^{-j\omega l}}$$

Example 3.16: Determine the frequency Response of

$$y(n) = x(n) + 0.8x(n-1) + 0.2x(n-2)$$

with $x(n) = e^{j\omega n}$

• **Solution:**

$$H(e^{j\omega})e^{j\omega n} = e^{j\omega n} + 0.8e^{j\omega(n-1)} + 0.2e^{j\omega(n-2)}$$

$$H(e^{j\omega})e^{j\omega n} = e^{j\omega n} + 0.8e^{-j\omega}e^{j\omega n} + 0.2e^{-j2\omega}e^{j\omega n}$$

$$H(e^{j\omega}) = 1 + 0.8e^{-j\omega} + 0.2e^{-j2\omega}$$

Homework 3.2:

- Determine the frequency Response of

$$y(n) = 0.2y(n-1) + x(n) + 0.8x(n-1) + 0.2x(n-2)$$

with $x(n) = e^{j\omega n}$

Example 3.17: Determine the output

$$y(n) = 0.8y(n-1) + x(n)$$

for $x(n) = \cos(0.05\pi n)u(n)$

• **Solution:**

$$H(e^{j\omega})e^{j\omega n} - 0.8H(e^{j\omega})e^{j\omega(n-1)} = e^{j\omega n}$$

$$H(e^{j\omega})e^{j\omega n} - 0.8H(e^{j\omega})e^{j\omega n}e^{-j\omega} = e^{j\omega n}$$

$$H(e^{j\omega}) = \frac{1}{1 - 0.8e^{-j\omega}}$$

For $x(n) = \cos(0.05\pi n)u(n)$

This means $\omega_0 = 0.05\pi$ therefore

$$\begin{aligned} H(e^{j0.05\pi}) &= \frac{1}{1 - 0.8e^{-j0.05\pi}} = \frac{1}{1 - 0.8 \cdot 1 \angle -j0.05\pi} \\ &= 4.0928 \cdot 1 \angle -0.5377 \\ &= 4.0928e^{-j0.5377} \end{aligned}$$

This means the system modifies the magnitude and phase of input by 4.0928 and -0.5377 radians respectively.

The output $y(n)$

- Now the output can be written as

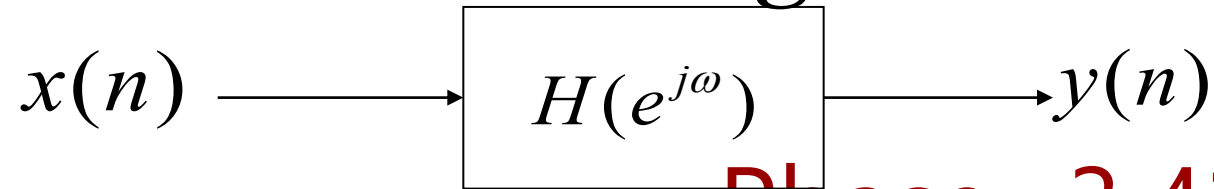
$$y(n) = 4.0928 \cos(0.05\pi n - 0.5377)$$

Frequency is the
same as the input

- or the phase can be written in terms of **number of samples** as

$$y(n) = 4.0928 \cos[0.05\pi(n - 3.42)]$$

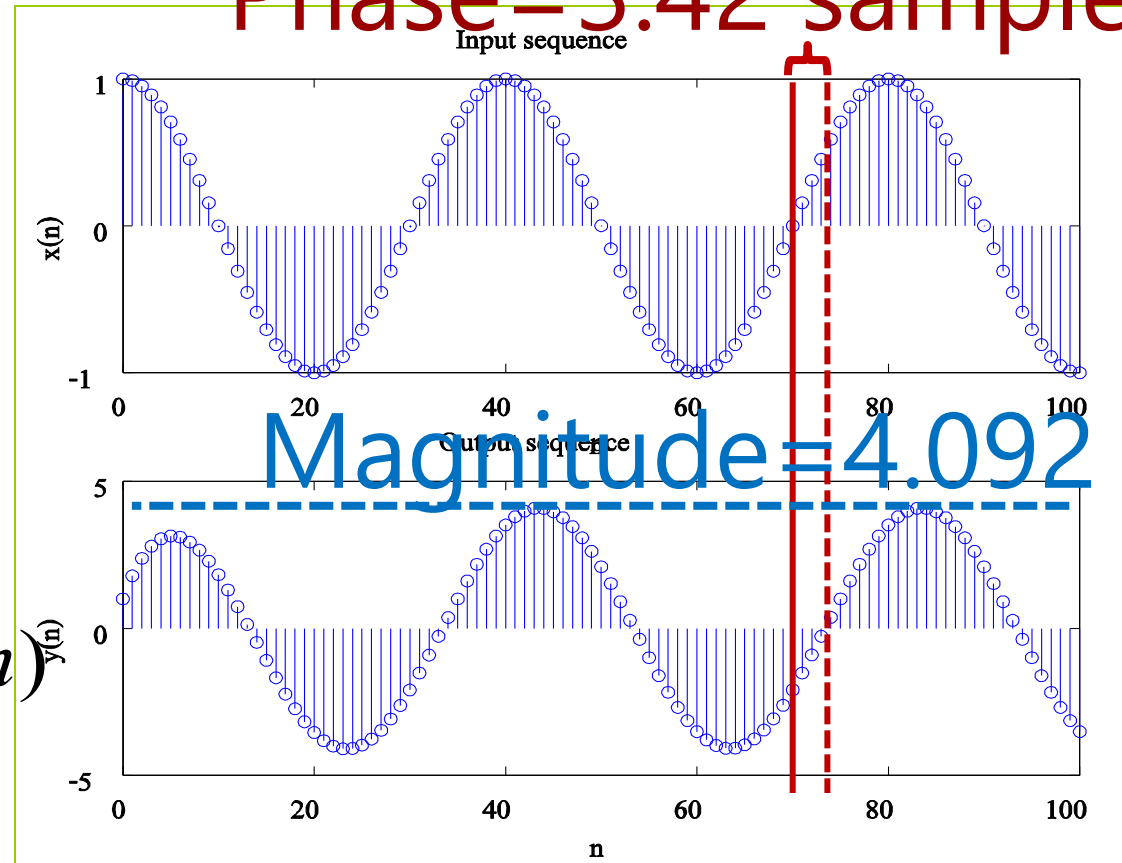
Testing



Phase=3.42 samples

$$x(n) = \cos(0.05\pi n)$$

$$y(n) = \left(\frac{1}{1 - 0.8e^{-j\omega}} \right) x(n)$$



Example 3.18: Determine the output

$$y(n) = 0.8y(n-1) + x(n) + 0.5x(n-1)$$

for $x(n) = \cos(0.05\pi n)u(n)$

• Solution:

$$y(n) - 0.8y(n-1) = x(n) + 0.5x(n-1)$$

$$H(e^{j\omega})e^{j\omega n} - 0.8H(e^{j\omega})e^{j\omega(n-1)} = e^{j\omega n} + 0.5e^{j\omega(n-1)}$$

$$H(e^{j\omega})e^{j\omega n} - 0.8H(e^{j\omega})e^{j\omega n}e^{-j\omega} = e^{j\omega n} + 0.5e^{j\omega n}e^{-j\omega}$$

$$H(e^{j\omega}) - 0.8H(e^{j\omega})e^{-j\omega} = 1 + 0.5e^{-j\omega}$$

$$H(e^{j\omega}) = \frac{1 + 0.5e^{-j\omega}}{1 - 0.8e^{-j\omega}}$$

For $x(n) = \cos(0.05\pi n)u(n)$

This means $\omega_0 = 0.05\pi$ therefore

$$\begin{aligned} H(e^{j0.05\pi}) &= \frac{1 + 0.5e^{-j0.05\pi}}{1 - 0.8e^{-j0.05\pi}} = \frac{2.57 \angle -0.03}{0.244 \angle -0.537} \\ &= 6.122e^{-j0.59} \end{aligned}$$

$$\begin{aligned} y(n) &= 6.122 \cos(0.05\pi n - 0.59) \\ &= 6.122 \cos[0.05\pi(n - 3.756)] \end{aligned}$$

Homework 3.3

- Consider the system with difference equation

$$y(n) = 0.8y(n-1) - 0.2y(n-2) + x(n) + 0.1x(n-1)$$

- Determine the output of the system when

$$x(n) = \cos(0.05\pi n)u(n)$$